

TABLE OF CONTENTS

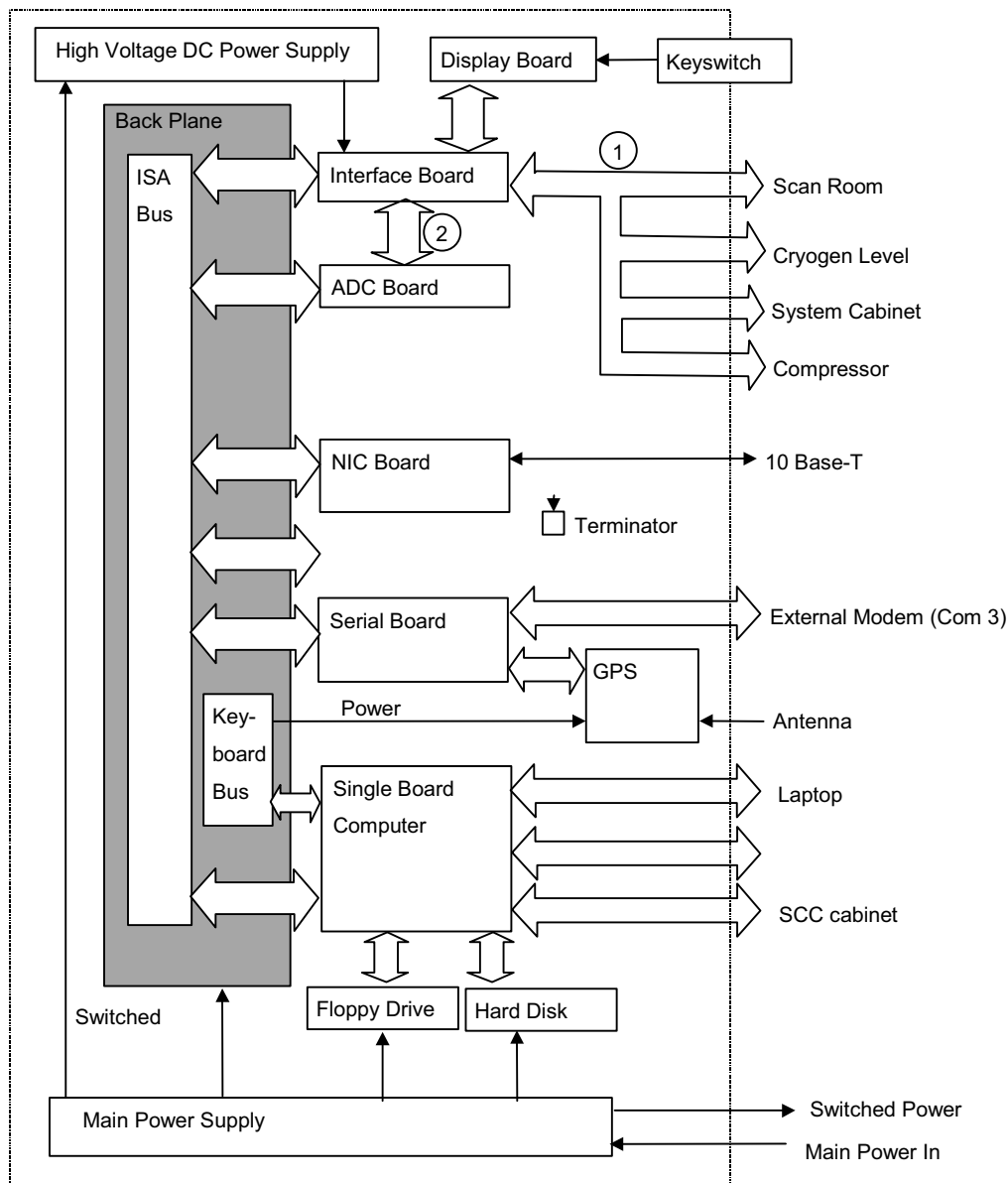
table of contents	1
1- Block Diagram and Theory	4
1-1 Magnet Monitor Internal Block Diagram	4
1-2 Scope	4
1-3 Reference Documents	5
1-4 System Requirements	5
1-5 Service Objectives	5
1-6 Magnet Monitor Overview	5
1-6-1 Monitored Elements and Features	6
1-6-2 Magnet Monitoring Processes	7
1-6-3 Sensor Data Collection	7
1-6-4 Data Transfer to PPADS	7
1-6-5 Alarm Notification	8
1-6-6 Fill Notification	8
1-6-7 Magnet Monitoring Components	9
1-6-8 MR Magnet Monitor Software	9
1-6-9 Water Flow and Temperature Sensors	10
1-6-10 Pressure Transducer	11
1-6-11 Reed Switch	12
1-6-12 Key Switch	12
1-6-13 RJ11 Multiplexer/Switch Box (optional)	12
1-6-14 User Interface	12
1-6-15 Compressor Diagnostic Port	12
1-6-16 Helium Probes	13
1-6-17 Helium Level Readings	13
1-6-18 Magnet Temperature Sensors	13
1-6-19 Remote Alarm box	13
1-7 Magnet Monitor Configuration	13
1-7-1 Left Side Panel	14
1-7-2 Connector Interface Panel	15
1-7-3 Interfaces to the Magnet and System Cabinet	15
1-7-4 Magnet Monitor for Stand-Alone Magnet	17
1-7-5 Data Collecting and Calculations	17
1-7-6 Other External Interfaces	19
1-8 SOFTWARE OVERVIEW	19
1-8-1 Proprietary Software	19
1-8-2 Non-Proprietary Software	20
1-8-3 Gracefull shutdown of the Magnet Monitor Software	20
2- Pro-Active Software Installation	21
2-1 Installing the Proprietary software	21
2-1-1 Connecting to the Magnet Monitor using a Laptop	22
2-1-2 Configuring the Magnet Monitor for HFO	22
2-1-3 Installing the software	23
2-1-4 Seting Time and Date	23
2-1-5 Restoring Insite Information	24
2-2 Online Center Install Check-out for Modem (See Section 2-3 for broadband)	24
2-3 Configuring magnet monitor for Beyond Modems	27
2-4 Verify Operation of Magnet Monitor	29
2-5 Subscribing to a Site	29
3- Connections	30
3-1 Overview	30
3-2 Local Connection – Laptop	30
3-3 Ethernet Network Connection – RJ-45	30
4- User Interface Screen Descriptions	32
4-1 Overview	32

4-2 Accessing the Pro-Active Software	33
4-2-1 Ethernet Connection	33
4-2-2 Laptop Connection	34
4-3 Status Screens	35
4-3-1 System Screen	35
4-3-2 Current Values Screen	36
4-4 Diagnose Screens	38
4-4-1 Logs Screen	38
4-4-2 Tables Screens	41
Magnet Heater Duty Cycle Graph (HDC)	46
4-4-4 Remote Screen	47
4-5 Configure Screens	48
4-5-1 Parameters Screen	48
4-5- 2 System Screen	50
4-5-3 Alarms Screen	51
Alarms Screen	51
4-5-4 Installation Check-out Screen	52
5- System Configuration	53
6- Parameter Setup	54
6-1 Regular Parameter Set-up	54
6-2 Coldhead Cycling	55
Individual Parameter Screen	56
7- Alarm Setup	57
7-1 Regular Alarm Setup	57
7-2 New alarms for HFO	58
7-3 Remote Alarm Activation	58
7-4 Alarm Activation	59
8- System Functional Tests	60
9- Troubleshooting	61
9-1 Troubleshooting tables	61
9-2 Boot-up Timeline	64
9-3 Software Structure	65
9-4 Files on Proprietary Disk	67
9-5 Dumping Magnet Monitor Data to a Floppy	67
9-6 Changing time and date	68
9-7 Determining Software Revision for the Magnet Monitor	68
9-8 Archiving Insite Information	69
10- Returning to Non-Proprietary Config	70
11- Renewal Parts	72
11-1 Magnet Monitor	72
11-2 Related Parts	72
12- Maintaining data for a particular magnet	73
13- GLOSSARY	74
Appendix A - Laptop Serial Connection Setup	75
Appendix B - Null Modem Dial-Up Setup for Laptop	76
B-1 Adding a Null Modem Connection	76
B-2 Configuring old style Laptop	77
B-2 Step 2 Configuring a DELL Laptop	78
Appendix C - Browser Setup	79
C-1 Microsoft Internet Explorer	79
C-2 Netscape Navigator 2.02	80
C-3 Netscape Navigator 4.01	80
C-4 Connecting your Browser to the Magnet Monitor	80
Appendix D - Coef.cfg File	82
Appendix E - Calculating with the coef.cfg File Data	84
Appendix F - General I/O Board Jumper Settings for later granite (post January 2001) units	87
APPENDIX G - PPADSCMD EMAIL	88
G-1 Overview	88

G-2 Receiving Magnet Monitoring Data Through Email	88
G-2-1 Address Email	89
G-2-2 PPADSCMD Guidelines	89
G-2-3 Use the HELP Command to See Correct Command Syntax	90
G-2-4 Use the BEGIN and END Commands to Prevent Incorrect Processing	91
G-3 Finding Site Specific Information	92
G-4 Receiving Information About All Magnet Monitoring Sites	95
G-5 Receiving Magnet Monitoring Program Information	98
G-6 Entering And Retrieving Team Contact Information	99
G-7 PPADSCMD Command Summary	100
revision history	101

1- BLOCK DIAGRAM AND THEORY

1-1 Magnet Monitor Internal Block Diagram



MAGNET MONITOR INTERNAL BLOCK DIAGRAM
ILLUSTRATION 1-1

1-2 Scope

This section contains specific information related to the Magnet Monitor, "Smart Helium Meter" (SHeM). The information contained herein, when used in addition to that contained in the referenced documents, will provide personnel with sufficient information to Operate, Install, Troubleshoot, and maintain a Magnet Monitor.

1-3 Reference Documents

2229090-100	Magnet Monitor Operators Manual
2261116	Magnet Monitor Hardware Installation Manual
2252032-3	Magnet Monitor Proprietary diskette, revision 2.3

1-4 System Requirements

The document references the Magnet Monitor for the HFO or Signa Openspeed magnet.

1-5 Service Objectives

The HFO magnet has a cryogen vessel that contains 340 liters of liquid helium, compared to 1,940 liters for an LCC magnet. It is imperative that the HFO system has a functional Magnet Monitor at all times. Magnet Monitoring has the following main objectives:

- Reduce cryogen consumption
- Provide proactive magnet servicing
- Provide remote access to magnet parameters to assist in problem diagnosis
- Improve cryogen fill schedule and efficiency
- Improve customer up-time

1-6 Magnet Monitor Overview

The Magnet Monitor (MM) is a stand-alone computer, with the exception of two signal inputs from the Systems Cabinet, and of course the signal connections to the Magnet. The Magnet Monitor is connected to a UPS system. The UPS will provide power to the Magnet Monitor up to 30 minutes after input power is lost. There is a "Remote Alarm" located near the operator's which informs the customer of a problem.

The Non-Proprietary version of the software provides a reduced set of functional operation. This reduced capability is identified in section 1-6-1, and will not be separated here. With regard to the MM hardware, there is no difference between the two modes, other than the fact that certain measurements are not taken, or controls and functions not activated for the Non-Proprietary case.

Magnet Monitoring consists of hardware and software used in conjunction with the Magnet Monitor to acquire, analyze, and store magnet subsystem data. This data is then transferred via the PPADS System to the various support centers, where it can be reviewed and acted upon.

This section contains the objectives of the Magnet Monitoring program, lists the elements that are monitored, describes the Magnet Monitoring processes, and provides details about the components in the Magnet Monitoring Subsystem.

The Magnet Monitor is microprocessor-based equipment, utilizing a standard PC type computer as a central processing component.

There are several general capabilities built into the unit, and two basic software modes. The Non-Proprietary Software mode provides a subset of the full capability of the Magnet Monitor. For the list below, "(Non-Pro)" indicates a function that is available in the Non-Proprietary mode. All features and capabilities are available in the Proprietary mode.

1-6-1 Monitored Elements and Features

The Magnet Monitoring equipment monitors, or controls, the following parameters:

- Measure and record system parameters
- Helium Level for upper and lower cryostat (Non-Pro)
- Up to 4 cryogenic temperatures
 - Recondenser (SI-410 diode)
 - Recondenser (RUO)
 - Shield (at recondensing coldhead)
 - Shield (at single stage coldhead)
- Helium vessel pressure (Non-Pro)
- Various Compressor parameters
 - Helium temperature high
 - Helium pressure low
 - Compressor on
 - 24 VDC present
 - Fuse off
 - Cooling water flow
 - Water temperature
 - Compressor run time
 - Heater on too long
 - Pressure recovery time after scanning
- Magnetic field detection
- Helium fill in progress
- RF present
- MR System Cabinet Powered up
- Magnet heater duty cycle
- Control certain Magnet properties/functions
 - Helium vessel heater (Non-Pro)
 - Automatic Compressor Reset (Non-Pro)
 - Manual Compressor Reset (via laptop)
 - Manual Helium sampling
 - Communications Test to PPADS
- Maintain internal data base
- Generate alarms for out-of-limit conditions
- Provide audible and visual alarm to customer for low helium condition
- Automatically call GE OnLine Center to report alarm conditions
- Receive calls from the OnLine Center and report status
- Allow local access with a laptop computer (GE Employees only)
- Allow local access on the Systems Console via Ethernet Connection
- Review measurement status
- View graphic or tabulated data
- Change parameter limits
- Generate local compressor reset
- Generate cryogen level reading
- Allow logging of critical notes or observations by GE Personnel

- Activate External Alarm for certain Parameter Failures

The Magnet Monitor is designed to facilitate certain other features, some of which are not currently implemented (indicated by “Future”). The following list shows those features;

- Web browser compatibility (TCP/IP & HTML)
- Global Positioning System (GPS) compatibility (Future)
- Interface to the Systems Cooling Cabinet, SCC (Future)

1-6-2 Magnet Monitoring Processes

The Magnet Monitoring Subsystem has 10 primary processes:

- Collect data from each of the sensors.
- Provide automatic compressor reset.
- Control helium vessel pressure.
- Notify the PPADS system when a parameter is outside of its acceptable limits.
- Notify customer via audible alarm when helium level is low.
- Notify the PPADS system when a fill is performed.
- Send data to the PPADS system to provide trend and historical data.
- Send data to host for control of Coldhead Cycling.
- Permit local and remote access to current and historic data.
- Permit performance test data transfer to the ASC using APA protocol.

1-6-3 Sensor Data Collection

The Magnet Monitoring Subsystem collects data from three areas.

- The magnet room, where sensors measure the vessel pressure as well as the first and second stage shield cooler temperatures and magnetic field.
- The system cabinet, where an RF Unblank signal and PDU Line Trigger signal are monitored.
- The shield cooler compressor, from which the shield cooler sensor monitors digital status signals (Leybold and Sumitomo compressors only) and measures the flow rate and temperature of the water going into the shield cooler.

1-6-4 Data Transfer to PPADS

After the Magnet Monitoring hardware and software is installed, the subsystem saves the parameter data and calls into the PPADS system daily to place the data in the database. Normal operating data, alarm information, and fill data are all transferred to the PPADS system. This data is gathered by the IRI system on the mainframe and is used for forecasting cryogen orders and deliveries as well as for developing trends of magnet performance.

You can receive this data by sending email to the PPADS System with the appropriate commands. See Appendix F for the PPADS email commands.

1-6-5 Alarm Notification

If a parameter goes outside of its low or high limits (alarm), then the Magnet Monitor contacts the PPADS System. The PPADS System transfers alarm data to the appropriate support center. For America's pole the OLC engineers are assigned the dispatch. In Europe and Asia the OLC engineers are notified of a magnet monitor alarm via e-mail. The engineers would review the data and take the necessary action.

An OLC MR engineer may take any of the following actions:

- Notify the customer that power needs to be restored to the equipment at the site.
- Reset the compressor remotely (if the site has a Sumitomo or Leybold compressor).
- Split the dispatch and assign it to an FE.
- Notify the customer about water supply problems.
- Disable an individual parameter alarm (Alarm Holdoff) for up to five days while parts are being ordered, the system is being serviced, or the parameter is recovering to normal condition.
- Request trend data from PPADS to see if a part is beginning to fail.
- Call a Field Engineer (FE) directly for more information.

Note

Anyone can receive notification of an alarm via e-mail, by subscribing to sites that have a Magnet Monitor. Reference Appendix F on PPADS of this manual to learn how to subscribe.

1-6-6 Fill Notification

If the subsystem detects a fill, then the Magnet Monitor contacts the PPADS System. The PPADS System transfers a copy of the fill data to IRI and also generates a fill report for engineers subscribed to that site.

1-6-7 Magnet Monitoring Components

The Magnet Monitoring Subsystem contains the following components:

Primary Components

- MR Magnet Monitor
- MR Magnet Monitor Non-Proprietary Software
- MR Magnet Monitor Proprietary Software
- RuO Buffer Amplifier
- UPS Power Supply
- Remote Alarm box
- Water Flow and Temperature Sensor
- Magnetic Reed Switch
- Key for Fill Switch

Auxiliary Components

- Compressor w/Diagnostic Port
- Helium Probe (p/o magnet subsystem)
- Pressure Transducer (p/o magnet subsystem)
- Shield Temperature Diodes (p/o magnet subsystem)
- Heater Element (p/o helium probe)

Optional Components

- RJ11 Multiplexer (Switch Box)
- Laptop Computer w/Windows 95 or higher
- Motorola Codex Modem (3265)

1-6-8 MR Magnet Monitor Software

The Magnet Monitoring software programs are loaded from a 3.5 inch diskette. The data collected by the programs is stored on an internal hard drive. This data is accessible locally using the console Ethernet connection, a laptop computer, or remotely via the OnLine Center. For further information regarding Magnet Monitor software, refer to those sections.

1-6-9 Water Flow and Temperature Sensors**Note**

For HFO magnets we are connecting the Water Flow and Temperature sensor to the input of the Cryo-cooler compressor, located on the right side of the SCC.

The Water Flow and Temperature Sensors are combined in one package. This device provides a proportional, linear electrical output for both parameters while providing a visual measurement for water flow.

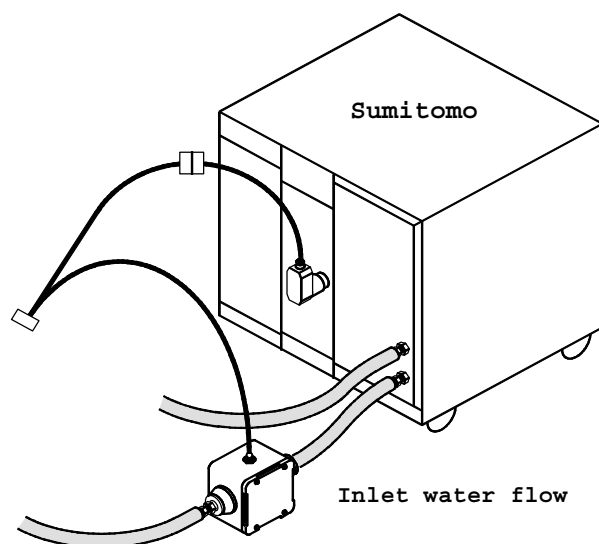
Specifications

Flow Range: 0-5 gpm or 6.6-17.0 LPM

Temperature Range: 0-100 Deg F or 4.5-24.0 Deg C

Pressure Rating: Up to 250 psi or 0.65-15.5 KPA

Accuracy: $\pm 4\%$



WATER FLOW AND TEMPERATURE SENSORS
ILLUSTRATION 1-3

TABLE 1-2
WATER FLOW AND TEMPERATURE SENSORS

No.	Column	Description
1	Compressor Interface Cable	For Sumitomo compressors via a 9 pin sub-D connector.
2	Water Flow & Temp sensor	Water flow is both electronic and visual. Water temperature is by thermocouple.
3	24" Hose	
4	Outlet Water Line	Using a ½ inch hose barb.

1-6-10 Pressure Transducer

The Pressure Transducer monitors the helium vessel pressure.

Specifications

Range:15 psig

Excitation:12 VDC

Output:1-5 VDC

This transducer aids in assessing:

- Proper maintenance of vessel pressures before, during, and after a helium fill
- Boil-off issues
- Magnet quench
- Pressure control for HFO magnets
- Data to the system for coldhead cycling

1-6-11 Reed Switch

The Magnetic Reed Switch is mounted on the bottom of the magnet. This switch provides an open contact when subjected to a magnetic field and a closed contact in the absence of a magnetic field. This switch helps the MR engineer determine if there is a loss of the magnetic field.

1-6-12 Key Switch

The Key Switch is mounted in the front panel of the Magnet Monitor. Field engineers and Air Products Technicians use this Switch when a helium fill is performed. Actuating the switch at the beginning of the fill causes the software to create and open a fill data file. This file stores the helium level, vessel pressure, and time/date. When the key is removed at the end of a fill, the file is sent to the PPADS system then on to the IBM Mainframe for cryogen administration purposes.

A key is distributed to each individual responsible for filling magnets. Return key to its storage area.

1-6-13 RJ11 Multiplexer/Switch Box (optional)

This multiplexer is used to switch between the Magnet Monitor Modem and the InLink Modem on the Signa Computer.

Note

The Magnet Monitor should always be connected to port 1 of the Mutliplexer.

1-6-14 User Interface

The User Interface allows the user to access the Login and Password prompts on the Magnet Monitor, as well as access to the Linux Operating System prompt. Login is required for access to the floppy drive as well as other software-supported features.

There are two methods of On-Site User Interface. First is the Console Interface connection. Second is Laptop connection. It is allowable for either User Interface to be connected, and logged in, while maintaining communications capability with the OLC.

The Laptop connection is accomplished using a Null Modem serial cable (9-pin female to 9-pin female) between the Laptop and the Magnet Monitor.

1-6-15 Compressor Diagnostic Port

The Sumitomo Compressor provides a Diagnostic Port. This port is used to provide the following status signals and remote function:

- Helium pressure low
- Helium gas temperature high
- Control module On/Off
- Compressor On/Off
- Remote compressor reset

1-6-16 Helium Probes

There are two helium probes for the HFO magnet. One is located in the upper vessel, the other is located in the lower vessel. The Helium Probes are mounted to the magnet cartridge inside the helium vessel. These probes act like a variable resistor and the output voltage is proportional to the resistance of the probe.

The probe must receive the appropriate drive current for proper operation. For the HFO magnet, the drive current is 75 milliamps. The constant current drive source is located in the Magnet Monitor. This current source is not field adjustable.

1-6-17 Helium Level Readings

The Magnet Monitor automatically takes a Helium Level reading every hour. Readings may be initiated manually by depressing the sample button located on the Magnet Monitor, or by activating a reading via the Laptop Magnet Monitor screens. The OLC can remotely sample the Helium Level.

1-6-18 Magnet Temperature Sensors

Ruthenium Oxide (RuO), and SI410 cryogenic sensors are located inside the magnet at various locations. These sensors each require a 10 microamp current source for excitation. There are four (4) such current sources within the Magnet Monitor. Each sensor is monitored every minute. The readings are available in the Current Value status screen. These current sources are not field adjustable. Verify accuracy by comparing readings to those of an independent instrument.

1-6-19 Remote Alarm box

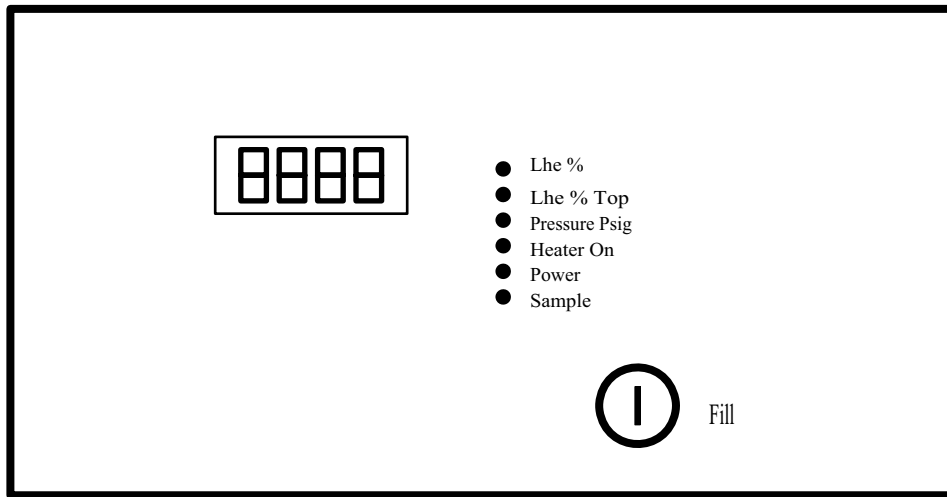
The remote alarm box is an Audible and visual indicator that is placed near the operator's work station. It informs the customer of an issue with the magnet. The remote alarm box will alarm under the following circumstances:

- RUO temperatures have been out of limits for greater than 90 minutes.
- Helium level for either vessel is out of limits
- Software for the magnet monitor has hung.

The customer can press a button on the Remote Alarm box to silence the audible alarm. There is a 6 hour timer that will re-activate the audible alarm if the alarm condition persists. The Remote Alarm box can also be disabled via the Diagnostics/Remote screen. See table 4-13 on how to disable the Remote Alarm.

1-7 Magnet Monitor Configuration

The Magnet Monitor is enclosed within an industrial type housing. Refer to Illustration 1-8 below. This unit weighs approximately 24 pounds, and is to be securely fastened to a wall located within the equipment room, in the orientation as shown below.



MAGNET MONITOR (TOP VIEW)
ILLUSTRATION 1-8

The local display system is comprised of a 4 digit 7-segment display, and several LEDs. The 7-segment display is for displaying Cryogen levels (in percent), and Helium vessel pressure (in PSIG). The LEDs indicate which parameter is being displayed at any given moment, with the exception of the “Heater” LED that indicates when the Helium vessel heater is activated.

The “Fill” switch is activated during the time cryogen fill is in progress. When in the fill mode, the cryogen levels are updated once per minute, otherwise this reading is taken automatically only once per hour.

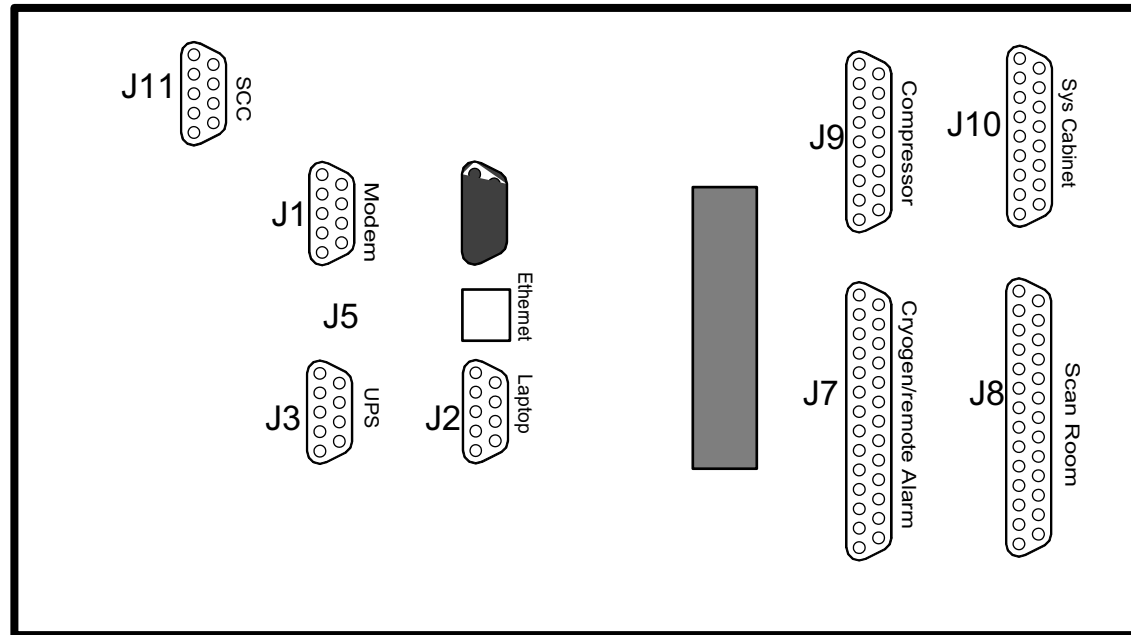
1-7-1 Left Side Panel

The left side panel of the Magnet Monitor contains the following.

- Power switch to power on the unit
- Reset button which will reset the unit
- Green HDD light, indicates when the internal hard drive is being accessed
- Floppy disk drive
- Floppy LED light, indicates when the floppy drive is being accessed

1-7-2 Connector Interface Panel

The lower right interface panel is used to interconnect to the Magnet subsystem, and other external interfaces. Refer to Illustration 1-10 below. Also refer to the Hardware Installation Manual for more specific connection information.



MAGNET MONITOR INTERFACE PANEL
ILLUSTRATION 1-10

1-7-3 Interfaces to the Magnet and System Cabinet

After installation, the 4 D-Sub connectors, J7, J8, J9, J10 provide all of the interface to the Magnet, System cabinet, and remote alarm box. For details on each connector, refer to the Hardware Installation Manual. Each pin is identified within that document.

System Cabinet Interface

Of these four connectors, one connects to the System cabinet, and picks up two signals there, which are utilized in the MM. One is the RF-Unblank signal from the IPG, and the other is a line trigger signal from the PDU.

The RF-Unblank is used to prevent the MM from activating a cryogen level reading during a scan, as well as to hold off making alarm decisions as long as a scan is in progress. The cryogen level readings are prevented due to the possibility that switching noise may create artifacts in the resultant images, and the alarm decisions are not made because the actual measured data during a scan may be corrupted by the RF or Gradient signals.

The line trigger signal is utilized as a means of determining if the System is powered up or not. Both of these signals, RF-Unblank and Line Trigger, are monitored and displayed in the MM status screen as well.

Compressor Interface

Another of the four connectors is routed to the Cryo-cooler compressor, here several status signals are monitored, and a reset line provides the ability for the MM to cause a reset of the compressor. Also, a water flow meter is attached to the cooling water outlet of the compressor. The water flow and its temperature are measured and displayed. The "Compressor Reset" signal and "Reset" control are the only two signals acted on. The other signals are monitored, displayed on the status screen, and logged for future reference. The details of these signals can be found in the Hardware Installation Manual.

Automatic Compressor Reset

The MM contains an automatic compressor reset feature. During normal operation, the MM monitors the "Compressor ON" signal from the cryogenic compressor. If this signal is found to be at a logic low (0 Vdc), a reset signal is activated to the compressor. If after a pre-set delay, the signal is still not present (meaning that the compressor did not respond favorably to the reset signal), another reset signal is initiated. This process is repeated at prescribed number of times. If after this time the compressor is still not indicating a "Compressor ON" condition, all further attempts to reset are stopped and the OnLine Center is notified via Modem that this fault condition exists. If, for some reason, the OnLine Center is not contacted (i.e. no phone connection), the alarm will be logged for future reference and the MM will continue normal operation. Whenever the MM recognizes that the compressor is once again operating, it will re-activate its ability to generate a reset signal again and clear the alarm condition.

Magnet Interfaces (Two Cables)

The two remaining interface cables are connected to the Magnet, via the Penetration wall. One of the cables carries the cryogen level monitoring signals, as well as the heater signal, which is used to maintain pressure for the HFO magnet. The Penetration wall is fitted with feed-through filters for both of these connections. The filters prevent extraneous noise leakage into the RF room.

Connector J7 has three separate outputs. Two are for helium level/magnet heater, the other is for the remote alarm box.

The cryogen level signal is produced by applying a current source of 75ma for the HFO magnet. A voltage is produced that is proportional to the cryogen level. Cryogen levels are displayed in percent (%) of full on the MM status screen. Except during an RF-Unblank period, cryogen levels are automatically sampled once per hour. Depressing the sample button will also initiate a new reading, and placing the Fill Key in the "FILL" position will cause a level reading update every minute.

The heater control signal is a 75ma current that is the same output as the helium level sensor supply. This signal is automatically activated whenever the magnet pressure drops below a prescribed limit, and is de-activated when the pressure crosses a prescribed upper limit. Generally these limits are set at 0.9 and 1.0 PSIG respectively. The heater output is applied to the lower helium vessel only.

Connector J8 is split four ways at the magnet. It connects to the pressure transducer, a magnetic reed switch and the coldhead sleeve (via a small "RuO Buffer amplifier" located on the top of the magnet), and the diode reading at the single stage coldhead.

A voltage is supplied to the pressure transducer, which in turn generates an output voltage proportional to the measured pressure; units of PSIG are used in the MM status screen.

The magnetic reed switch is located at the rear pedestal, it is closed when the magnetic field is down, and open when the field is up. This condition is sensed and displayed in the MM status screen. This is used to monitor for a quenched magnet condition.

The Preamp connector monitors the RUO temperature of the recondenser, the SI-410 diode at the recondenser, and the SI-410 diode located on the shield (near the dual stage coldhead). The other connector monitors the SI-410 diode that is located near the single stage coldhead.

In all four cases, the measurements are a 4-wire configuration. Two wires conduct a stimulus current of 10 microamps to each sensor, and the other two wires sense the developed voltage. For the two RuO channels, the developed signal is of a small magnitude (10mv to 25mv). For this reason the RuO Buffer amplifier is used to amplify these signals by a factor of 65. This results in an output range of 0.65 to 1.625 volts. Internally to the Magnet Monitor an additional gain of 2 is provided which increases these signals to the 1.3 to 3.3 volt range prior to application to the A/D converter.

The other two channels, used for SI410 or DT500 sensors, do not require further amplification.

1-7-4 Magnet Monitor for Stand-Alone Magnet

The system cabinet signals are not absolutely required for the MM to operate. In their absence the MM will operate with disregard to RF-Unblank. The PDU line trigger signal is used to notify the MM that RF-Unblank must be considered.

This allows the use of the MM in cases where it is necessary to monitor and control a magnet that is either separate from an MR system, or where the system is powered off.

1-7-5 Data Collecting and Calculations

The following list of parameters are currently capable of being monitored by the MM.

TABLE 1-4
PARAMETERS MONITORED BY MAGNET MONITOR

No.	Description	Sampling Rate	Units	Comments
1	Water Flow	1/minute	GPM	Parameter can be edited to use Liters per Minute (LPM).
2	Water Temperature	1/minute	F	Parameter can be edited to use Degrees F or C
3	Helium Vessel Pressure	1/minute	PSIG	
4	Compressor Status Signals (Multiple)	1/minute	N/A	
5	RF-Unblank	1/minute	N/A	
6	Line Trigger	1/minute	N/A	
7	Cryogen Level (Helium)	1/hour	%	Except during Fill, or when Sample button pressed
8	Cryogenic Temperatures (up to 4)	1/minute	K	
9	Magnetic Field detect (Reed switch)	1/minute	N/A	
10	Compressor Run Time	1/minute		Can view percentage of "on" time for a given month.
11	Heater Duty Cycle			Can plot percentage of time magnet heater was on for a 48 hour period.
12	Pressure recovery	2 hours	PSIG	Will determine recovery time of pressure which will monitor cryo cooler performance

Each parameter measured, not including the LOGIC type signals, is converted from a voltage into their respective units of measure. This is accomplished in software, by applying the reading to a pre-determined polynomial, having selectable coefficients, depending on the reading type.

The coefficients are stored in a configuration file (coef.cfg) (see Appendices D and E), and have been pre-determined for each sensor type. Each measured parameter can have up to 5 polynomials, with 5 coefficients, to provide the curve fitting necessary for accurate conversion. These coefficients will not require alteration unless a sensor type is changed to a different type, having a different transfer function. It is not recommended that field personnel alter these values in any way. To do so can cause erroneous readings, and defeats the need to maintain a common configuration among systems.

One anticipated need to alter these coefficients might be to display the readings in different units of measure. It is possible to create these additions, however qualified personnel must create them and be common with regard to software configuration management.

While in the Proprietary mode of operation, a database of readings is maintained within the MM. For each parameter Minute, Hour and Day values are retained, up to a maximum of 2048 data points per type. For example, Water Flow will have 3 data files associated with it (Minute, Hour and Day), and so on for each other parameter. This data is used for table and graphic displays in the available screens, and is also accessible to the OnLine Center as an FTPed file.

1-7-6 Other External Interfaces

The Magnet Monitor is designed to provide three methods of external communication. Those are as follows:

- External Modem
- Serial Port
- Ethernet

External Modem Serial Port

The Magnet Monitor is equipped with an external modem (known as the Global Modem) for phone line connections. In this case, the modem is connected to the Magnet Monitor via a serial port, and the customer phone line is connected to the external modem. The phone line connection should pass through the UPS for transient protection.

Serial Port(s)

Aside from the serial port used for the external modem, there are three other serial interfaces. Two are labeled "Console" and "Laptop". Both are identical interfaces, and can be used interchangeably. Using a Null Modem serial cable, a Laptop computer can be connected here. These ports are configured for both PPP and TCP/IP connections, as well as normal serial RS-232 serial protocols. The final serial connection is a 486 connection and will be connected to the Systems Cooling Cabinet (SCC).

The user can configure their Laptop as a Hyperlink terminal, or as a Null Modem Dial-up terminal. In the Hyperlink terminal mode, using the provided Linux Logins and passwords, it is possible to perform standard Linux commands, and thus communicate with the MM. A typical use of this method would be to view files, directories and so on. Using a Dial-up session, while logged in as a PPP user, the connection will allow the use of an Internet browser, such as Netscape or Microsoft Internet Explorer. Using a browser you can view and interact with the HTML screens provided by the Magnet Monitor.

Ethernet Interfaces

There is an Ethernet connection between the Magnet Monitor and the host. This connection is needed to communicate vessel pressure and helium data to the host for Coldhead Cycling. The connection is 10 Base 2. Service will be able to access the Magnet Monitor via the Ethernet to view the HTML screens, and to access the software.

1-8 SOFTWARE OVERVIEW

1-8-1 Proprietary Software

The MM is preloaded with a copy of Proprietary software for initial release. Its default configuration is set for operation with an HFO magnet type.

In the Proprietary mode, the MM is capable of the following functions:

- Measure and display locally and via HTML Cryogen pressure
- Measure and display locally and via HTML Cryogen levels
- Measure and display (via HTML) water flow

- Measure and display (via HTML) water temperature
- Measure and display (via HTML) up to 4 cryogenic temperatures
- Measure and display (via HTML) Compressor status
- Measure and display (via HTML) System Cabinet Status (RF-Unblank and Trigger)
- Monitor and display (via HTML) Magnet field status (Ramped/Quenched)
- Allow certain restricted and UN-restricted Logins via the serial port, using a laptop computer
- Provide tabulated and graphic displays of collected data (Min, Hour & Day) (HTML)
- Facilitate local "Lamp Test", to verify local display is functional
- Maintain a local user log, user entered events and notes
- Control a heater used to regulate magnet pressure
- Monitor and provide automatic compressor reset
- Communicate via Ethernet connection (Future)
- Allow the use of an external or internal Modem
- Report all out-of-limit alarms to the OLC
- Allow remote compressor reset
- Maintain log of alarm events
- Support FTP file transfers
- Respond to OLC queries
- Facilitate FILL mode
- Maintain compressor Run Time

1-8-2 Non-Proprietary Software

In the Non-Proprietary mode, the MM is capable of the following functions;

- Measure and display locally Cryogen pressure
- Measure and display locally Cryogen levels
- Control a heater used to regulate magnet pressure
- Monitor and provide automatic compressor reset
- Provide data to host for coldhead cycling

1-8-3 Gracefull shutdown of the Magnet Monitor Software

To properly reset, or turn power off to the magnet monitor, do the following:

- Hold the *Sample* button in for about four seconds. **halt** should appear on the display.
- Release the *Sample* button for less than 1 second, and then press the *Sample* button again. **oooo** (four lower case o's) should appear on the display.
- Wait for at least **90 seconds**, then you may reset or power down the Magnet Monitor.

2- PRO-ACTIVE SOFTWARE INSTALLATION

2-1 Installing the Proprietary software

1. At the Login prompt, type in **root**

At the password prompt, type in **mvserve**

Note

Some of the early Granite magnet monitors had a different login and password name. If the above login and password does not work try:

Login **ac79-E38** (case sensitive password)

Password **RvmL&Z2pFc** (case sensitive password)

2. Insert the Proprietary software floppy disk into the floppy drive on the Magnet Monitor.



VERIFY THAT YOU HAVE THE PROPER PROPRIETARY DISKETTE INSTALLED. FAILURE TO USE THE PROPER DISK WILL DESTROY THE HARD DRIVE ON THE MAGNET MONITOR. OBSERVE THE REVISION OF NON-PROPRIETARY SOFTWARE PRIOR TO LOADING PROPRIETARY SOFTWARE. THE FOLLOWING INFORMATION WILL APPEAR AFTER CONNECTING THE MAGNET MONITOR SERIAL PORT TO YOUR LAPTOP, AND ENTERING HYPERTERMINAL.

If the revision shown is: Magnet Monitor Non-Proprietary Software Rev. 2.3 Red Hat Linux release 6.1 (Cartman) Kernel Rev. 2.2.12-20

Then the Proprietary software that should be loaded is: **2252032-3 yellow floppy**

If the revision shown is: Magnet Monitor Non-Proprietary Software Rev. 2.0 Red Hat Linux release 6.1 (Cartman) Kernel Rev. 2.2.12-20

Then the Proprietary software that should be loaded is: **2252032-2 yellow floppy**

If the revision shown is: Magnet Monitor Non-Proprietary Software, Rev. 1.3 Red Hat Linux release 4.1 (Vanderbilt) Kernel Rev. 2.0.27

Then the Proprietary software that should be loaded is: **2252032 red floppy**

2-1-1 Connecting to the Magnet Monitor using a Laptop

1. Using a serial connection (9-pin female to 9-pin female null modem), and a laptop computer, open a Hyperterminal window (Hyperlink connection, refer to section 5 in the Hardware Installation Manual 2230681). If you do not have a copy of direction 2230681, do the following:

On the laptop in Windows:

- a. Select Start/Programs/Accessories/Communications/Hyperterminal
- b. Click on Hyperterminal
- c. Create a new connection named magnet monitor set-up
- d. Select direct to Com1, then OK
- e. Edit Bits per second to 115200, data bits to 8, parity to none Then OK
- f. Hit the enter key to get the Login prompt

Note

Some laptops may not have the hyperterminal executable available on their laptops. You will need to configure your laptop by doing the following steps:

Select Start-> Settings-> Control Panel-> Add/Remove Programs.

Select Windows setup on the top of the menu.

Select Communications, Details, Hyper Terminal, OK

Select Apply which is located on the lower right area of the window.

Select Start -> Find-> Files or Folders and type in Hypertrm, look in C:

Double click on Hypertrm.exe to begin Hyperterminal program.

2-1-2 Configuring the Magnet Monitor for HFO

The Magnet Monitor should have default settings for the HFO system. To verify the settings, connect the laptop to the Magnet Monitor as in step 2-1-2.

At the login prompt type: **setup**

Case sensitive password

At the password prompt type: **monitor**

Case sensitive password

The following will appear on the screen:

Do you wish to change the Configuration Type [y/n] **Y**

There are 2 choices for Configuration Type:

1. hfo
2. lcc

Enter your choice as a number from 1 to 2, or Q to quit: **1 [enter]**

If needed, you can change the Compressor Reset Values. The values mention below should not be changed.

Do you wish to change the Compressor Configuration values? [y/n] **Y**

Do you wish to Enable the Compressor Reset Function? [y/n] **Y**

Enter value for Startup Delay Time (minutes) [1:5] **5**

Enter value for Reset Delay Time (minutes) [1:30] **10**

Enter value for Recheck Delay Time (hours) [1:5] **1**

Enter value for Retry Count (count) [0:10] **2**

Will this Magnet Monitor be serially connected to a UPS [y, n] Should be **Y** for all HFO systems, the UPS is part of the system.

The Magnet Monitor will now reset.

2-1-3 Installing the software

2. Once the Magnet Monitor has reset hit [Enter] on the laptop.

3. At the Login prompt, type in **root**

At the password prompt, type in **mvserve**

4. Insert the proper Proprietary software floppy disk, part number into the floppy drive on the Magnet Monitor.

5. Mount the drive by typing the following:

mount -t msdos /dev/fd0 /mnt/floppy (case sensitive, don't ignore spaces) Press [Enter].

6. Copy files to /tmp directory. **cp /mnt/floppy/* /tmp** Press [Enter].

6. After the copy function is completed, unmount the floppy drive.

umount /dev/fd0 Press [Enter].

7. Remove floppy disk from drive.

8. Execute the install script. **/tmp/installp** Press [Enter].

9. When the install script completes, the Magnet Monitor will re-boot. This will cause a disconnect in your laptop connection.

2-1-4 Setting Time and Date

1. Once the Magnet Monitor has reset hit [Enter] on the laptop.

2. At the Login prompt, type in **root**

At the password prompt, type in **mvserve**

3. At the prompt type in **date MMDDhhmmyy** where

MM is the 2 digit month

DD is the 2 digit day

hh is the 2 digit hour in military time

mm is the 2 digit minute

yy is the 2 digit year

Example is **date 0816132001** for August 16, 2001 at 1:20pm.

4. To write the date into firmware type **/sbin/clock --systohc** If you omit this step the magnet monitor will loose the time and date when rebooted.
5. Type **reboot** to reset the Magnet Monitor.

2-1-5 Restoring Insite Information

If you are replacing a magnet monitor and have previously archived the Insite information, you can restore it at this time.

1. Install the **Insite Information** floppy.
2. Re-connect using Hyperlink terminal again. Login as **root**, password **mvserve**.
3. Mount the floppy by typing **mount -t msdos /dev/fd0 /mnt/floppy** [enter]
4. Type **/MVServe/bin/check-out** [enter]

The laptop screen will echo back that it is about to re-load the checkout information.

Type **yes** to continue.

The program will pull the information from the floppy onto the Magnet Monitor hard drive.

It will take a few seconds for the data to transfer.

The laptop screen will then prompt you to "hit return to continue". Depress any key on the keyboard at this time. After another few minutes the Magnet Monitor will re-boot. The Magnet Monitor will then perform in the same state as the original.

IMPORTANT

If you have successfully completed Insite Information Restoration, do not run InitConfig. It will destroy all check-out information. Run communication test to verify connection with PPADS.

2-2 Online Center Install Check-out for Modem (See Section 2-3 for broadband)

The following steps will establish check-out with the OLC.

1. After the Magnet Monitor re-boots(re-booting can take up to 3 minutes), re-connect using the Hyperlink terminal again. Login as **root**, password **mvserve**.
2. On your laptop, via the terminal window, type in the following command to initiate a script file: **cd /MVServe/bin**

3. Press [Enter].
4. **./InitConfig**
5. Press [Enter]. Answer all questions. The following is a list of typical answers:

Do you want to Continue with Configuring Magnet Monitor?

***** WARNING!!! *****

If yes, you will remove any previous Insite Checkout information.

Enter (Y/N): y

Hospital Name: Putra Memorial

'Putra Memorial' correct? (Y/N) y

Dial Prefix:'1' correct? (Y/N) y You may need to enter 9,1 if the phone line at
your site requires a 9 before the 800 number**Tone or Pulse Dialing? (T/P):** t

't' correct? (Y/N) y

Custom Modem Initialization

String: <enter modem string here if required> return

'<enter modem string here if required>' correct? (Y/N) y

Select a modem type from the list:

(1) External_Motorola_Codex

(2) Internal_US_Robotics_336

(3) InitConfigCustom

(4) Robotectics33

(5) Internal_US_Robotics_288

(6) Internal_Hayes_Accura_144

(7) Generic

(8) Generic_288

(9) External_US_Robotics_288

Pick selection (1-9): **7 for the Global Modem**

OK

Use current Country? US (Y/N) Select **N** if outside the States, and go through the list until you find the correct country.

Modem Initialization string

String 'AT &FO X4 &DO EO T MO Q2 SO=1 S7=90'

Do you want to use default string (Y/N) Select **Y** for the default or **N** if you want to change the default. The default should work for the majority of countries.

Do you want to continue with configuring Magnet Monitor's network?

*****Warning!*****

If yes, you will remove any previous network configuration

Enter (Y/N) **Y**

Magnet Monitor network address 3.44.56.78

Use default value (Y/N) **N**

Enter new IP address for Magnet Monitor **3.87.23.54** Example of network address
 provided by Hospital Network
 Administrator

Netmask 255.255.252.0

Use default value (Y/N) **Y**

Host IP address 3.44.56.78

Use default value (Y/N) **N**

Enter new IP address for Host **3.87.23.53** Example of network address
 provided by Hospital Network
 Administrator

Magnet Monitor config changed.

Machine will reboot in one minute.

6. The Magnet Monitor is now ready for Insite installation (On-Line Center).

Contact the Global Service Support Center in your GEMS service pole and work with your Pole OLC Magnet Support Group within the Service Support Center to complete an Magnet Monitor Check-out (1-800-321-7937 in GEMS Am). You will be required to provide the following information.

- Site/Hospital name
- GE System ID and system type for customer system connectivity via the Magnet Monitor Computer (This information is obtained from the OLC)
- IP address of the magnet monitor
- Magnet Serial number
- MAC Team leader
- Modem phone number and MUX channel if applicable.

They will place a call to your Magnet Monitor and initiate a file transfer. This file is sclink. When this file transfer ends, and Configlink is executed automatically, your Magnet Monitor will re-boot once again.

After the OLC has completed the check out the Ethernet connection between the Magnet Monitor and the Console will be activated. You can then view the status and the settings of the Magnet Monitor on the console. Refer to Section 3-3 on how to connect to the Magnet Monitor. A backup method of connecting to the Magnet Monitor is to use your laptop via a Dial-up connection and Null Modem cable. Refer to Appendix B and C for these setups.

Once your browser is functional, select the Configuration screen, Alarms, and activate the Master Alarm (set Enable to yes) if all parameters are operating correctly. Be sure to select the "SAVE" button to save this change. The Magnet Monitor is now ready to automatically report alarms to the OLC. Refer to section 4 through 7 for proper set-up.

Note

Allow 24 hours after InSite Check out before running Communications Test
(Refer to Illustration 4-15.)

2-3 Configuring magnet monitor for Beyond Modems**Do you want to Continue with Configuring Magnet Monitor?**

***** WARNING!!! *****

If yes, you will remove any previous Insite Checkout information.

Enter (Y/N): y

Current Connectivity Configuration: 'Modem'

Use default value? (Y/N) **N**

Select from the following configurations:

(0) Not Configured

(1) Modem

(2) Beyond Modems

2

Beyond Modem Selected

'/etc/ppp/options.ttyS2.PRECHK': to present used to replace '/etc/ppp/options.ttyS2'!

'/etc/hosts.PRECHK': to present used to replace '/etc/hosts'!

'/etc/sysconfig/network.PRECHK': to present used to replace '/etc/sysconfig/network'!

'/etc/ppp/ip-up.PRECHK': to present used to replace '/etc/ppp/ip-up'!

Hospital Name: 'Institut Mutualiste Montsouris'

Use default value? (Y/N) **N**

Enter proper hospital name

Do you want to Continue with Configuring Magnet Monitor's Network?

***** WARNING!!! *****

'Y' to continue, 'N' to exit, 'X' to remove network config.

Enter (Y/N/X): **y**

MagMon IP Address: 10.250.24.91 (**Magnet Monitor IP Address**)

Enter proper Magnet Monitor Address

Netmask: 255.255.255.128 (**Magnet Monitor Net mask**)

'255.255.255.128' correct? (Y/N) **Enter proper MM Netmask**

Gateway: 10.250.24.126 (**Router IP address**)

'10.250.24.126' correct? (Y/N) **Enter proper Gateway**

Host IP Address: 10.250.24.20 (**LX IP address**)
'10.250.24.20' correct? (Y/N) **Enter proper Host Address**

FE Laptop Address: 10.250.24.6 (**FE Laptop Ip address**)
'10.250.24.6' correct? (Y/N) y

Attention: After checkout the Magnet Monitor will accept IP connection only for these 3 IP address, all others will be refused!

Magnet Monitor Config changed
Changes take effect after next reboot.

Magnet Monitor ready for 'Check Out'.
Machine will reboot in One Minute

Broadcast message from root (ttyS1) Fri Jul 19 12:13:10 2002...

The system is going DOWN for reboot in 1 minute !!

Broadcast message from root (ttyS1) Fri Jul 19 12:14:10 2002...

Contact the Global Service Support Center in your GEMS service pole and work with the Pole OLC Magnet Support Group within the Service Support Center to complete an InSite Check-out (1-800-321-7937 in GEMS Am). You will be required to provide the following information.

- Site/Hospital name
- GE/HCS System ID and system type for customer system connectivity via the Magnet Monitor Computer (This information is obtained from the OLC)
- The manufacturer of this scanner
- Magnet Serial number
- MAC Team leader
- The model and computer (if required) of this scanner
- Modem phone number

They will place a call to your Magnet Monitor and initiate a file transfer. This transfer is sclink. When this file transfer ends, and Configlink is executed (by the OLC), your Magnet Monitor will re-boot once again.

Using a Dial-up connection and Null Modem cable, you may now use your laptop Internet browser to connect to the Magnet Monitor and view its current status and settings. Refer to Appendix B and C for these setups.

Once your browser is functional, select the Configuration screen, Alarms, and activate the Master Alarm (set Enable to yes). Be sure to select the "SAVE" button to save this change. The Magnet Monitor is now ready to automatically report alarms to the OLC. Refer to section 4 through 7 for proper set-up.

2-4 Verify Operation of Magnet Monitor

1. Verify that the parameters are properly configured in the table located in Section 6-1.
2. Need to put the Magnet Monitor into a fault condition to verify proper connection to the OLC by doing the following steps:
 - a. Verify that Magnet Monitor has been Entitled for at least 48 hours (it takes a while for the database to update).
 - b. Force the Magnet Monitor into ALARM condition by setting the upper limit of the vessel pressure alarm to 0.85psig.
 - c. Add a note in the "Comments" field indicating this is a test of the Magnet Monitoring system. Include a phone number where the OLC can contact you.
 - d. Wait for acknowledgement from OLC that the test was a success. If you do not receive acknowledgement, contact OLC to determine cause.
 - e. Verify that the upper limit for vessel pressure is returned to 6.25psig.

2-5 Subscribing to a Site

It is recommended that primary Field Engineers subscribe to their sites via E-mail. You will then be notified on E-mail if your site alarms for any reason, has a cryogen fill, or does a check-in with the PPADS database.

1. Go into Microsoft Exchange E-mail. Address E-mail to PPADSCMD.
2. In the body of the message type the following:
begin
subscribe 215341shem email (*where 215341shem is the cryo system ID of the site you wish to subscribe to*)
end
3. For more information on PPADSCMD mail server commands, refer to Appendix F.

3- CONNECTIONS

3-1 Overview

There are two ways to access the Pro-Active software on the Magnet Monitor:

- Through a local serial connection using a laptop computer. (Refer to Appendices A, B and C.)
- Through an Ethernet network using the RJ-45 connector for 10-baseT networks. This communication will be activated when the OLC successfully completes the Magnet Monitor check out procedure.

3-2 Local Connection – Laptop

Connect the serial connector on the laptop computer to the “Laptop” connector, J2 on the Magnet Monitor. Refer to Appendix A for setup information. Refer to Appendix B for dial-up settings.

Note

The “Hyperlink” connect type used in Appendix A is intended for installs and access to the Magnet Monitor operating system. The “Dial-up” and “Browser” described in Appendices B and C are used for local viewing of the MM HTML screens via a web browser.

IMPORTANT

If you need to reset, or power down the Magnet Monitor, do the following steps to properly shut down the software.

- Hold the *Sample* button in for about four seconds. **halt** should appear on the display.
- Release the *Sample* button for less than 1 second, and then press the *Sample* button again. **oooo** (four lower case o's) should appear on the display.
- Wait for at least **90 seconds**, then you may reset or power down the Magnet Monitor.

3-3 Ethernet Network Connection – RJ-45

The Ethernet connection exists between the Magnet Monitor and the Host after successful check out from the OLC. The magnet monitor can be accessed from the Common Service Desktop from Utilities menu. Select the option labeled **Magnet Monitor**. The following process can also be used.

1. Boot up the console to Signa level.
2. To execute communication, open a C-shell and type **su -** password is **operator**

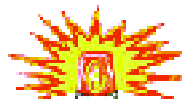
3. At the prompt, type **insite_browser -n**
4. Select **edit** from the Netscape menu, then select **preferences**.
5. Click on the arrow that is on the left side of the "Advanced" menu tree.
6. Select **Proxies** from the Advanced menu tree, then select **Manual Proxy Configuration**.
7. Select **View**, then click in the **No Proxy for** text box.
8. Delete any data that is in the text box and type **MAGNET_MONITOR_IP**
9. Click **OK** to exit the Pro Dialog Box, then click **OK** to exit the preferences Dialog Box.
10. Select **File** from the Netscape menu, then select **open page**
11. Enter **MAGNET_MONITOR_IP** in the text box, and then select **open**
12. You should then be able to log into the magnet monitor.
13. Login name is **MMService**, password is **MagnetMonitor**

The Pro-Active software has three basic types of screens:

- Status screens which show currently set values and limits of monitored parameters and their alarm status.
- Diagnostic screens which provide tests for troubleshooting the Magnet Monitor.
- Configuration screens which are used to set up the Magnet Monitor and change limits for the monitored parameters.

The Pro-Active screens are organized like a table with the main screen type names as column headings and screens within a screen type as row headings. The selected screen is displayed in the “table cell” part. The screens are selected by clicking on the screen names listed in the column and row headings.

The upper left hand corner area has an alarm indicator. If any monitored parameter has its alarm enabled and an alarm condition is detected, the alarm indicator changes from a green check mark to a red flashing alarm light.



ALARM INDICATOR
ILLUSTRATION 4-2



CHECK INDICATOR
ILLUSTRATION 4-3

4-2 Accessing the Pro-Active Software

4-2-1 Ethernet Connection

14. Boot up the console to Signa level.
15. To execute communication, open a C-shell and type **su** - password is **operator**
16. At the prompt, type **insite_browser -n**
17. Select **edit** from the Netscape menu, then select **preferences**.
18. Double click on the arrow that is on the left side of the "Advanced" sub tree.
19. Select **Proxies** from the Advanced sub tree, then select **Manual Proxy Configuration**.
20. Select **View**, then click in the **No Proxy for** text box.
21. Delete any data that is in the text box and type **MAGNET_MONITOR_IP**
22. Click **OK** to exit the sub tree, then click **OK** to exit the preferences sub tree.
23. Select **File** from the Netscape menu, then select **open**
24. Enter **MAGNET_MONITOR_IP** in the text box, and then select **open**
25. You should then be able to log into the magnet monitor.
26. Login name is **MMSERVICE**, password is **MagnetMonitor**

4-2-2 Laptop Connection

1. If you are using a laptop, connect it to the Magnet Monitor as described in “Local Connection – Laptop”, section 3.2, turn the laptop on and allow it to boot up. Otherwise, skip to the next step. Refer to Appendices A, B, and C for specific setup information.
2. Open the browser.
4. Click on [Open] and enter the Magnet Monitor’s IP address (10.0.0.5).

Login name is **MMService**, password is **MagnetMonitor**. Both are case sensitive.

4-3 Status Screens

4-3-1 System Screen

Status		Diagnose	Configure
System Current Values	Hospital Name:	bay 14b	1
	USN:	0000466718	2
	System Phone No.:	47362	3
	System Type:	Lightning	
	Magnet Type:	nMR	
	Magnet Serial Number:		
	Shield Cooler Type:	Sumitomo	
	MAC Team Leader:		
	Primary FE:		
	Software Revision Info:	Magnet Monitor Proprietary Software Rev 2.0 Red Hat Linux release 6.1 (Cartman) Kernel Rev. 2.2.12-20	
System Time:	Fri May 25 15:08:37 2001		
Last Check-in:	Mon Apr 30 13:57:42 2001		
Last Fill:	Sat Aug 12 11:00:33 2000	7	
Last Alarm:	Fri May 25 15:06:31 2001		

SYSTEM STATUS SCREEN
ILLUSTRATION 4-4

TABLE 4-1
SYSTEM STATUS SCREEN

No.	Field	Description
1	Site Name	Name of the hospital or geographic location.
2	System ID	A unique name assigned to each specific unit.
3	System Phone No.	Phone number for PPADS; should not change.
4	System Time	Current time of the time zone where the unit is installed.
5	Last Check-in	Name of the periodic check-in log file.
6	Last Fill	Name of the fill log file.
7	Last Alarm	Name of the alarm log file.

4-3-2 Current Values Screen

System

Current Values

1	2	3	4	5
Item Name	Value	Units	Alarm On/Off	Status
Water Flow	0.3675	GPM	Y	OK
Water Temp.	0	Deg F	Y	OK
Shield Temperature	48.49	K	Y	OK
Recondenser (SI-410) Temp	11.37	K	Y	OK
Recondenser (RUO) Temp	10.71	K	Y	OK
Coldhead (RUO) Temp	10.71	K	Y	OK
He Vessel Pressure	1.805	PSIG	Y	OK
He 2 (upper)	95.67	%	Y	OK
He 1 (lower)	98.86	%	Y	OK
Pressure Control	Htr Off	PSIG	N	OK
Helium Loss / Week	0	%/Wk	N	Off

Digital Inputs 6

System Cabinet On	0
Unblank	1
Magnet Quench Detected	1
Sample Cryogen PushButton	0
Fill Key Switch	0
Audible Alarm Enabled	0
He Meter Current Sensing Avail	1
Compressor +24VDC	0
Compressor Fuse Off	0
Compressor On	0
Compressor LHe Temp OK?	0
Compressor LHe Pres OK?	0
Klixon Error	0
Level Meter 1 Current	0
Level Meter 2 Current	0

Digital Outputs 7

Compressor Reset	0
Pressure Heater Enable	0

CURRENT VALUES SCREEN
ILLUSTRATION 4-5TABLE 4-2
CURRENT VALUES SCREEN

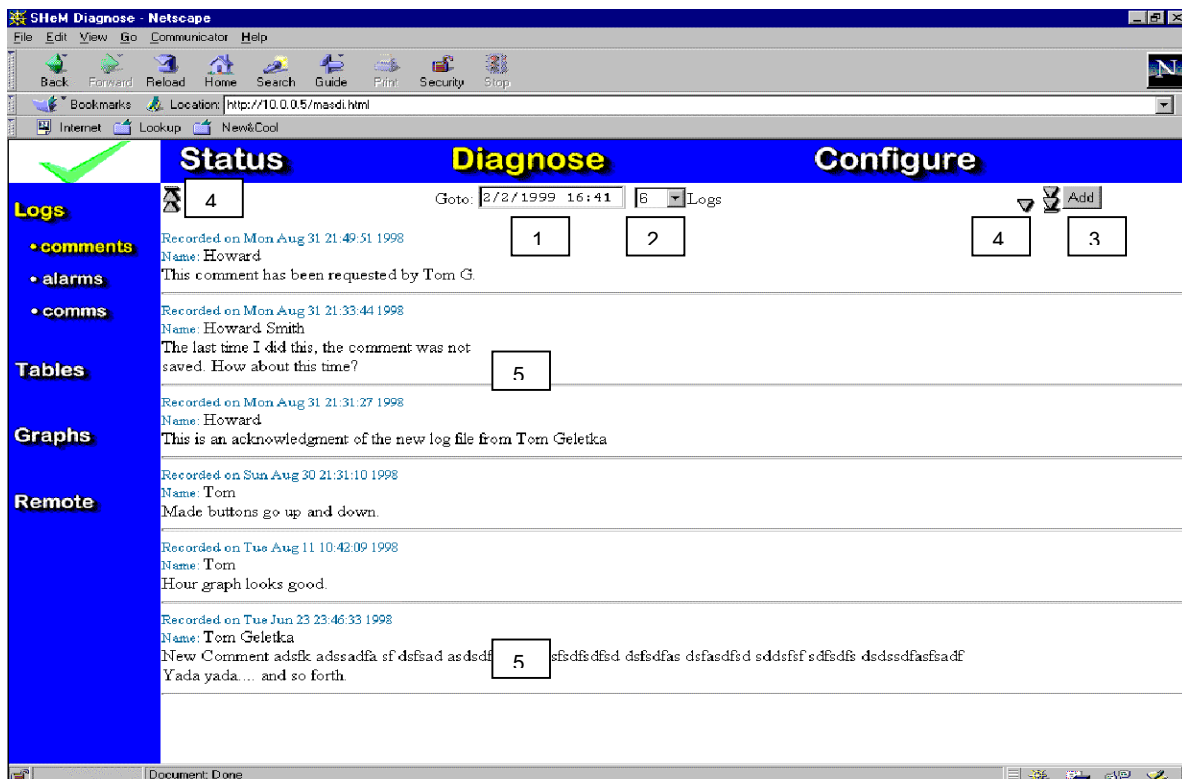
No.	Column	Description
1	Item Name	Name of the monitored parameter.
2	Value	Current monitored value.
3	Units	Units of measurement.
4	Alarm On/Off	Alarm On/Off setting. The display shows Y when the alarm for the monitored parameter is turned on and N when the alarm is turned off.
5	Status	Alarm status. The display shows HI or LOW when the parameter is in alarm state and OK when not in alarm state. Note: If the alarm is not enabled, OK will always be displayed.
6	Digital Input	Logic status of digital inputs. See table 4-3 for definitions of the status inputs.
7	Digital Output	Logic status of output functions.

TABLE 4-3
DESCRIPTION OF DIGITAL INPUTS

Digital Input	Value	Description
System cabinet on	1	System cabinet has power
RF unblank	1	System is scanning or rf unblank cable is off, which indicates a logic low.
Magnet Quench Detected	1	Magnet reed switch is closed, no magnetic field detected. Magnetic reed switch is open when properly in the magnetic field.
Sample cryogen pushbutton	1	The sample button was activated
Fill key switch	1	Magnet fill is activated
Audible alarm enabled	0	Normally off, if state changes to 1, the audible alarm has been activated for the customer
Helium meter current sensing available	1	This magnet monitor is capable of sensing the proper output current to sample cryogens.
Compressor 24VDC	1	Compressor has 24VDC
Compressor fuse off	1	Compressor fuse is OK
Compressor on	1	Compressor is running
Helium temperature high	1	Compressor has shut down due to high helium temperature
Helium pressure low	1	Compressor has shut down due to low helium pressure
Klixon Error		Not used at this time
Level Meter 1 Current	1	Magnet Monitor has detected an error in the current being used to read helium level. The helium level may be inaccurate for the lower vessel. Zero indicates a properly functioning helium meter.
Level Meter 2 Current	1	Magnet Monitor has detected an error in the current being used to read helium level. The helium level may be inaccurate for the upper vessel. Zero indicates a properly functioning helium meter.
Digital outputs		
Compressor reset	1	Compressor has been remotely reset
Pressure heater enable	1	Magnet heater is activated
Interrupt Enable		
Helium Source Current Enable	0	Activates when magnet monitor is outputting current
Remote Alarm Enable	0	Activates when remote alarm condition is enabled
Helium Current Pulse	0	Control bit for strobing helium meter current
Watchdog monitor	1	Verifies hardware and software monitor control is being sensed.
Helium meter update	0	Control bit for enabling helium meter current

4-4 Diagnose Screens

4-4-1 Logs Screen

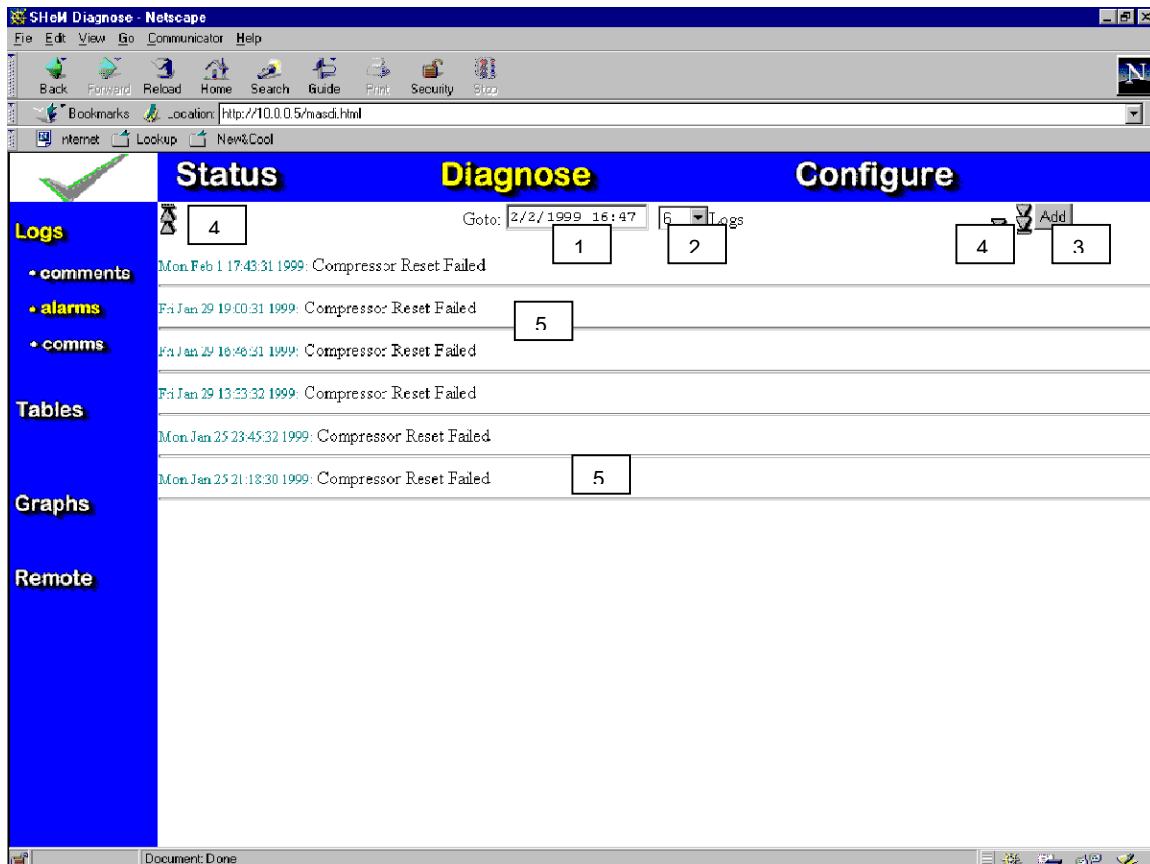


LOGS SCREEN
ILLUSTRATION 4-6

TABLE 4-4
LOGS SCREEN

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Logs	Select to make greater/lesser logs available.
3	Add	Select to add a logged entry.
4	Up/Down Arrows	As needed to scroll log entry.
5	Records	Actual example records.

Alarms Screen

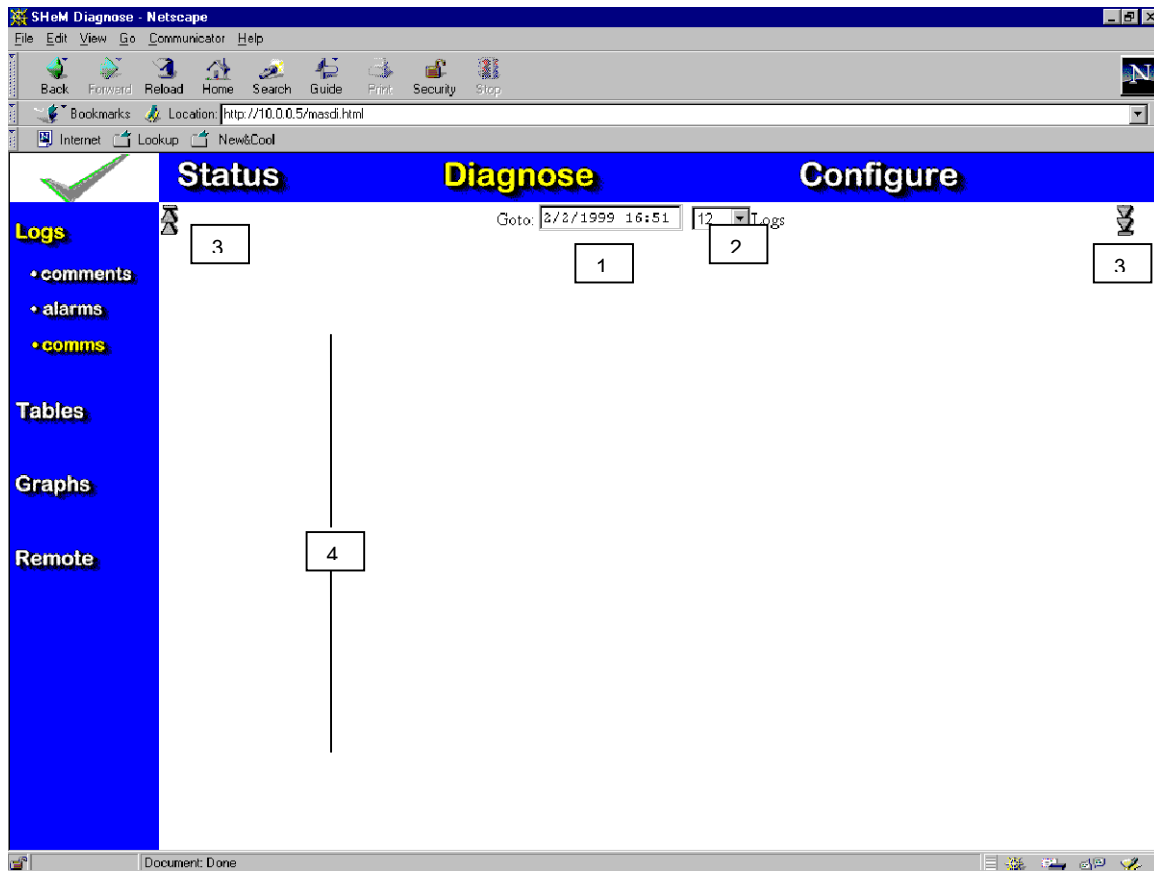


ALARMS SCREEN
ILLUSTRATION 4-7

TABLE 4-5
ALARMS SCREEN

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Logs	Select to make greater/lesser logs available.
3	Add	Select to add a logged entry.
4	Up/Down Arrows	As needed to scroll log entry.
5	Alarms	Recorded alarms.

Comms Screen



COMMS SCREEN
ILLUSTRATION 4-8

TABLE 4-6
COMMS SCREEN

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Logs	Select to make greater/lesser logs available.
3	Up/Down Arrows	As needed to scroll log entry.
4	Comm Information	Logged communications entries.

4-4-2 Tables Screens

Minute Screen

SHem Diagnose - Netscape

File Edit View Go Communicator Help

Back Forward Reload Home Search Guide Print Security Stop

Bookmarks Location: http://3.45.82.86/masdi.html

Internet New and Cool Look Up

Status Diagnose Configure

Goto: 6/19/2000 11:19 6 Cols.

Item Name	Units	6/19 11:18	6/19 11:17	6/19 11:16	6/19 11:15	6/19 11:14	6/19 11:13
Water Flow		0.4	0.2	0.2	0.6	0.0	0.2
Water Temp.	Deg F	0.0	0.0	0.0	0.0	0.0	0.0
Shield Temperature	K	573.04	591.91	580.04	49.62	49.99	47.73
Recondenser (SI-410) Temp	K	13.79	13.46	12.18	8.38	7.54	6.61
Recondenser (RUO) Temp	K	32.49	105.23	24.00	8.32	7.43	6.88
Coldhead (RUO) Temp	K	10.39	10.84	8.84	8.36	7.60	6.47
He Vessel Pressure	PSIG	1.94	2.01	1.98	1.97	1.92	1.92
He 2 (upper)	%	95.0	95.0	95.0	95.0	95.0	95.0
He 1 (lower)	%	98.9	98.9	98.9	98.9	98.9	98.9
Pressure Control	PSIG	0.00	0.00	0.00	0.00	0.00	0.00
Helium Loss / Week	%/Wk	0.00	0.00	0.00	0.00	0.00	0.00

Document: Done

MINUTE SCREEN
ILLUSTRATION 4-9

TABLE 4-7
MINUTE SCREEN

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Columns	Select to make greater/lesser columns of data available.
3	Right/left Arrows	As needed to scroll log entry.
4	Item Name	Parameter description.
5	Units	Units of measure.
6	Date/Time	Date and time data recorded. Digital signals can be viewed by double clicking on the date/time stamp.

Hour/Day Screen

Status **Diagnose** 1 2 **Configure**

Goto: 6/19/2000 11:21 4 Cols.

Item Name	Units	6/19/2000 10:00			6/19/2000 9:00			6/19/2000 8:00			6/19/2000 7:00		
		Low	Avg	High	Low	Avg	High	Low	Avg	High	Low	Avg	High
Water Flow	GPM	0.0	0.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Water Temp.	Deg F	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shield Temperature	K	32.72	204.10	591.91	32.35	33.58	35.00	32.72	33.82	34.61	32.35	33.80	35.00
Recondenser (SI-410) Temp	K	4.73	13.35	88.72	4.73	4.77	4.83	4.73	4.78	4.83	4.73	4.77	4.83
Recondenser (RUO) Temp	K	3.81	16.56	105.32	3.78	4.20	4.70	3.77	4.20	4.72	3.77	4.27	4.76
Coldhead (RUO) Temp	K	3.37	15.33	105.45	3.33	4.40	6.61	3.28	4.24	6.37	3.29	4.37	6.47
He Vessel Pressure	PSIG	1.25	1.61	1.89	1.27	1.38	1.60	1.27	1.41	1.51	1.28	1.45	1.62
He 2 (upper)	%	95.0	95.7	95.7	95.7	95.7	95.7	95.7	95.8	95.8	95.7	95.7	95.8
He 1 (lower)	%	98.9	98.9	98.9	98.9	98.9	98.9	98.9	98.9	98.9	98.9	98.9	98.9
Pressure Control	PSIG	0.00	0.20	1.00	0.00	0.10	1.00	0.00	0.00	0.00	0.00	0.25	1.00
Helium Loss / Week	%/Wk	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

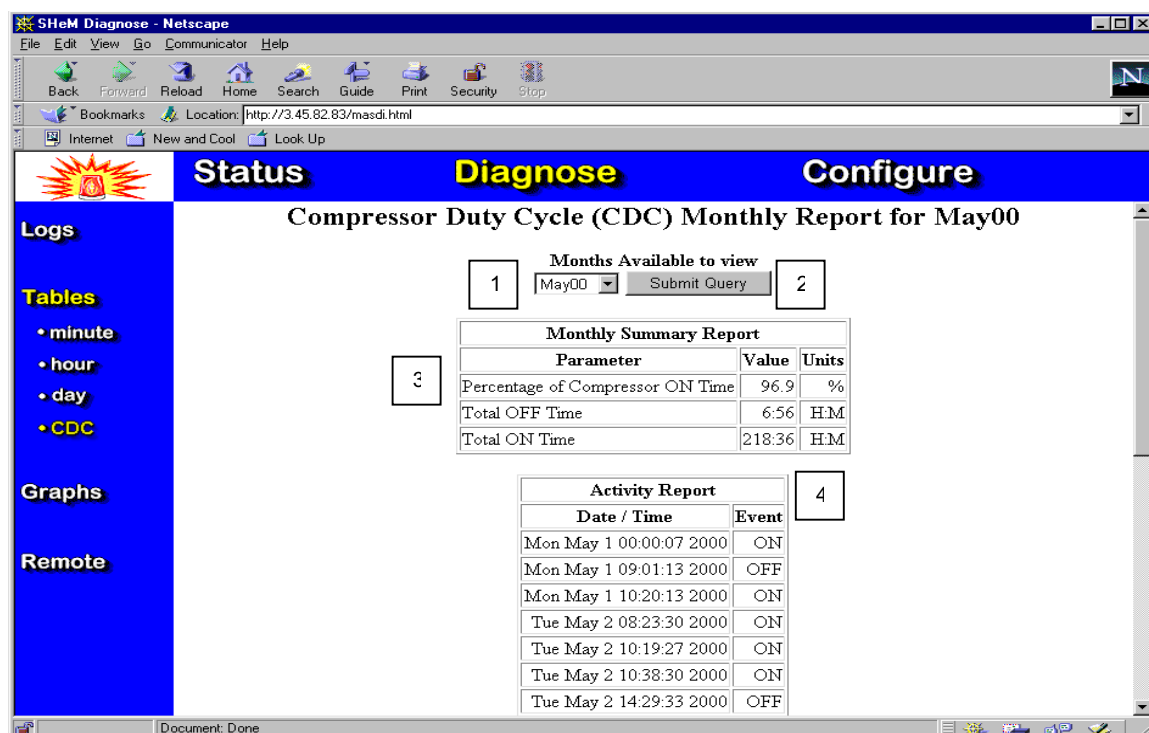
HOUR/DAY SCREEN

ILLUSTRATION 4-10

TABLE 4-8
HOUR/DAY SCREEN

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Column	Select to make greater/lesser column available.
3	Right/Left Arrows	As needed to scroll column entry.
4	Item Name	Parameter description.
5	Units	Units of measure.
6	Date/Time	Date and time data recorded. Digital signals can be viewed by double clicking on the date/time stamp.

Compressor Duty Cycle Screen



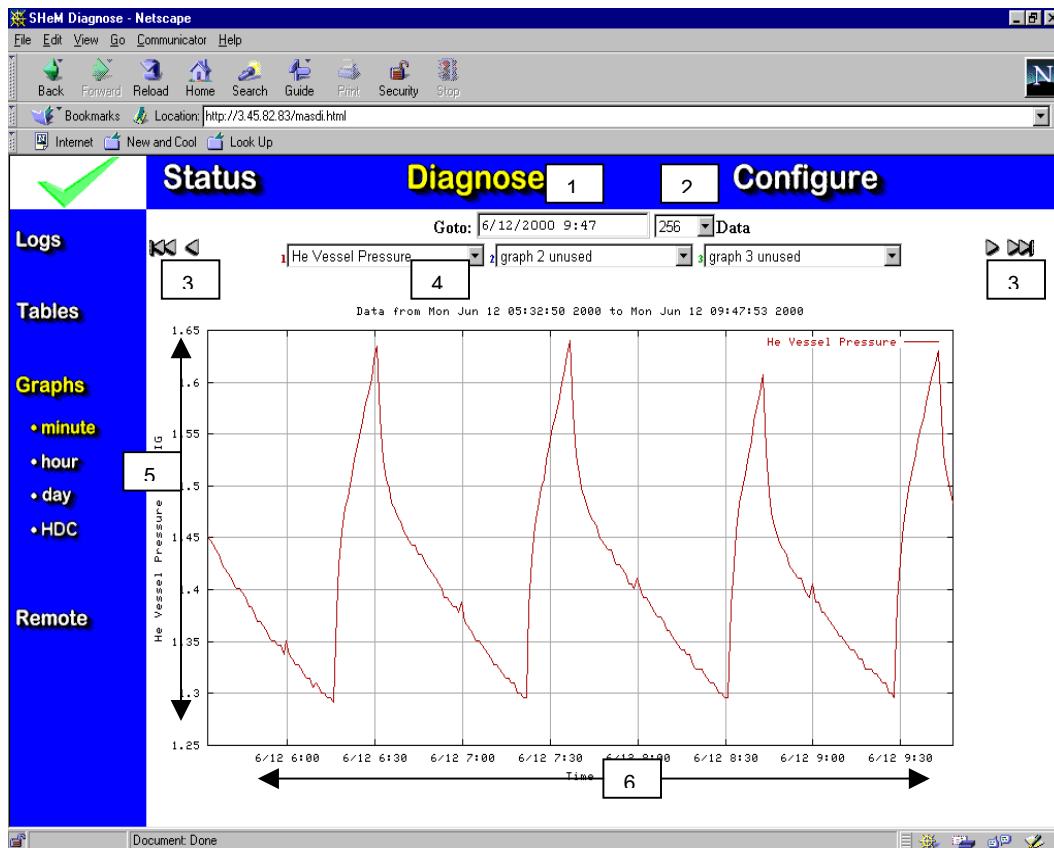
COMPRESSOR DUTY CYCLE SCREEN
ILLUSTRATION 4-12

TABLE 4-10
COMPRESSOR DUTY CYCLE SCREEN

No.	Field	Description
1	Month	Selects the month you wish to query.
2	Submit Query	Must be selected to activate query.
3	Percentage of compressor on time	Displays the percentage of time the compressor has been on for the month that was selected.
4	Activity report	Displays exact times the compressor was turned on or off. Two consecutive "ON or OFFS" means the Magnet Monitor was reset.

4-4-3 Graph Screens

Minute Graph

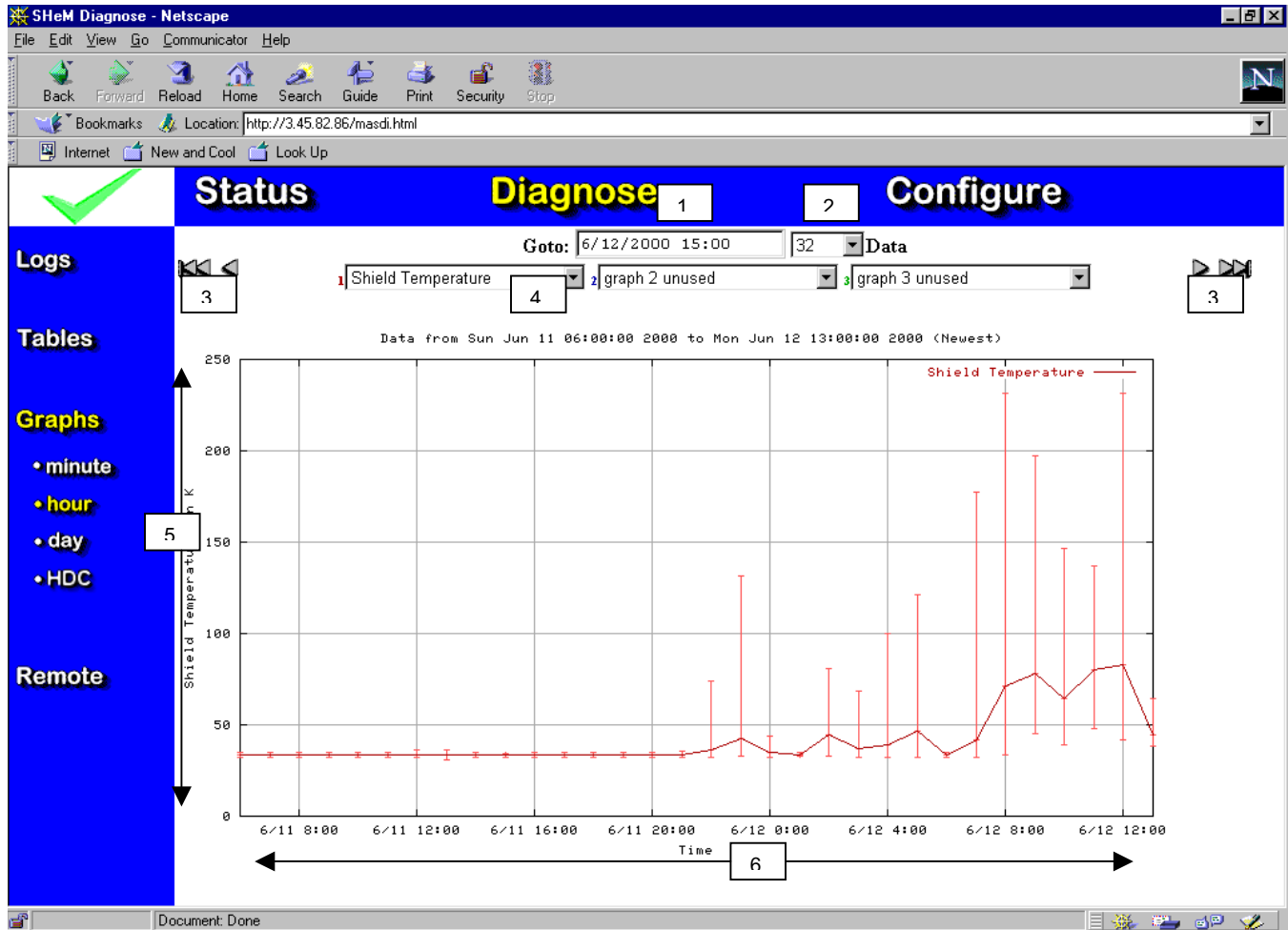


MINUTE GRAPH
ILLUSTRATION 4-12

TABLE 4-10
MINUTE GRAPH

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Data	Number of data points displayed.
3	Up/Down Arrows	As needed to scroll curve.
4	Graph Types	Parameter graph (up to three curves).
5	Magnitude Axis	Data magnitude.
6	Time Axis	Date and time of data.

Hour/Day Graph

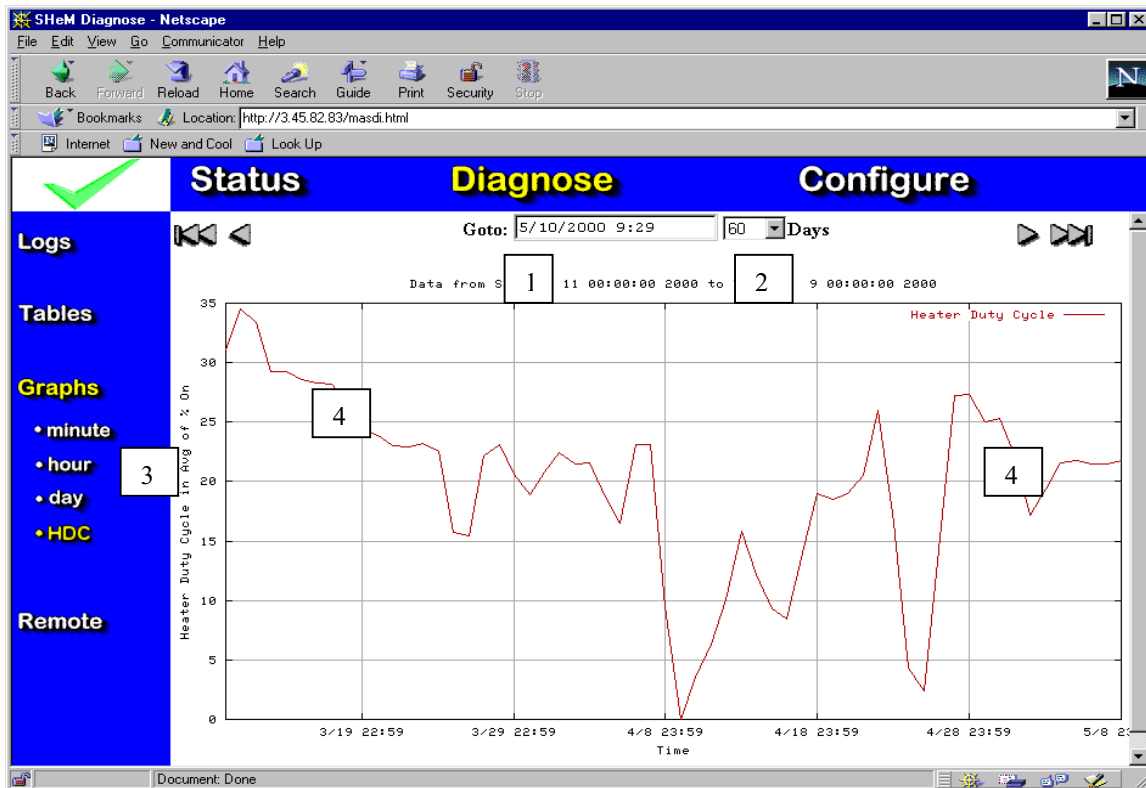


HOURLY/DAY GRAPH
ILLUSTRATION 4-13

TABLE 4-11
HOURLY/DAY GRAPH

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Data	Can change number of datapoints displayed.
3	Right/left Arrows	As needed to scroll curve.
4	Graph Types	Parameter graph (up to three curves).
5	Magnitude Axis	Data magnitude. For the Hour and Day data the bold line is the average of the 60 minute data points. The top tick mark is the highest data point for the one hour period, the bottom tick mark is the lowest data point for the one hour period.
6	Time Axis	Date and time of data.

Magnet Heater Duty Cycle Graph (HDC)

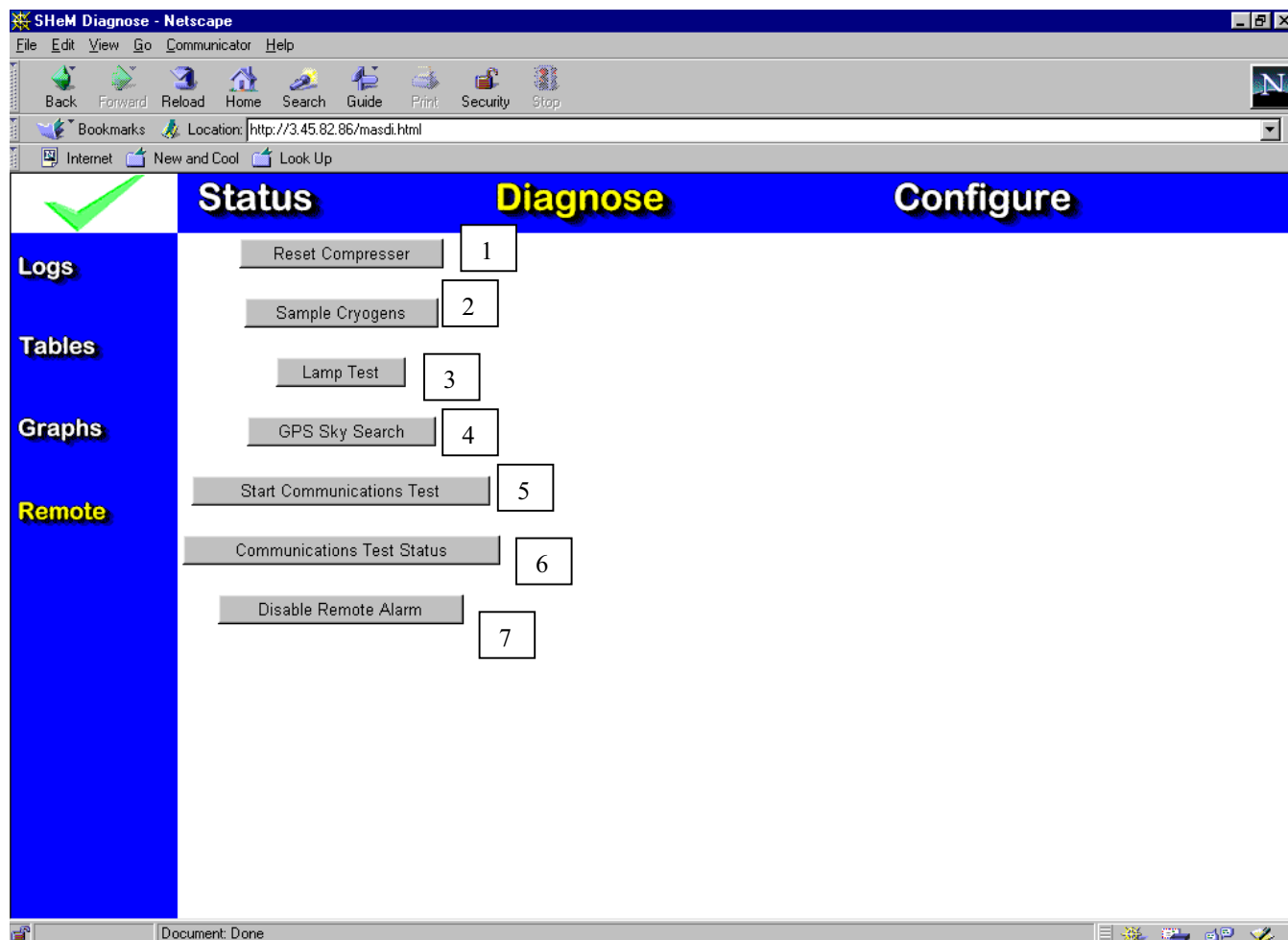


HEATER DUTY CYCLE GRAPH
ILLUSTRATION 4-14

TABLE 4-12
HEATER DUTY CYCLE GRAPH

No.	Field	Description
1	Date	Select and type in to view particular start date for duty cycle plot.
2	Number of days	Select the number of days you wish to plot
3	Heater Duty Cycle	Percentage of time the magnet heater is on. Data is averaged over a 48 hour period.
4	Plot of Duty Cycle	Actual plot.

4-4-4 Remote Screen



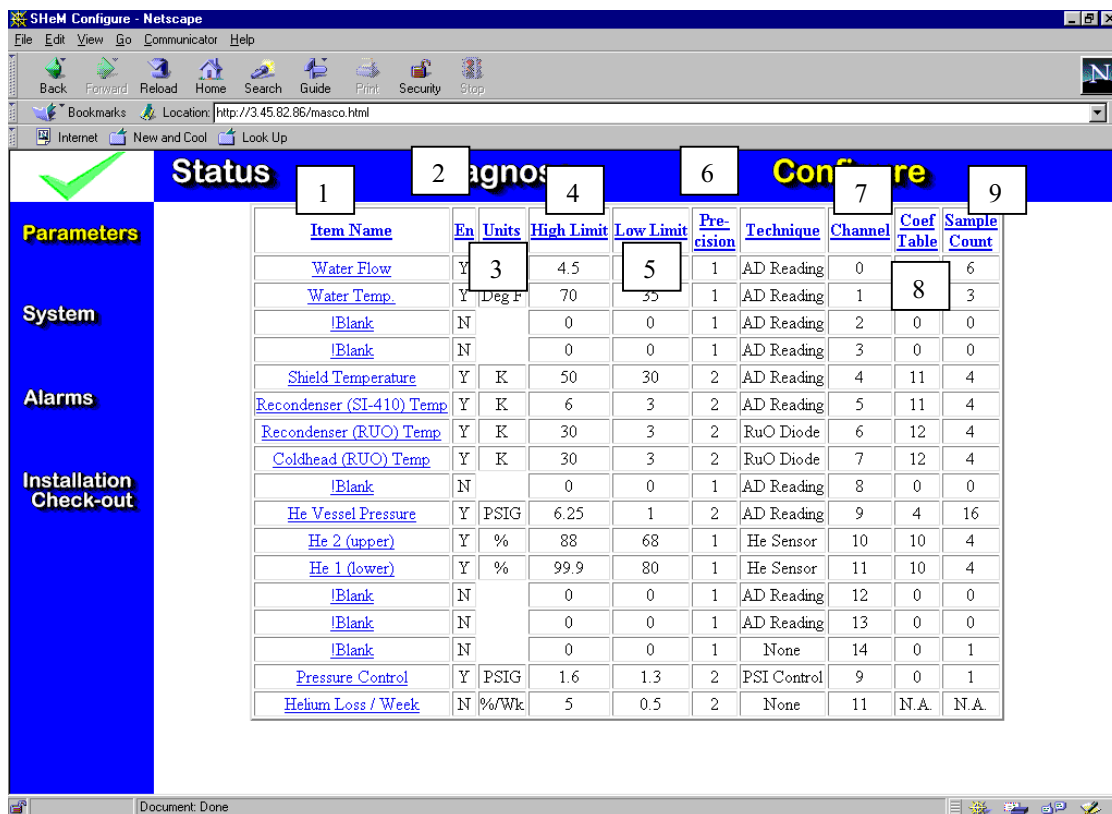
REMOTE SCREEN
ILLUSTRATION 4-15

TABLE 4-13
REMOTE SCREEN

No.	Button	Description
1	[Reset Compressor]	Resets compressor from support center location.
2	[Sample Cryogenics]	Initiates a cryogen sample from support center location.
3	[Lamp Test]	Activates MM local display (all segments).
4	[GPS Sky Search]	Not currently functional.
5	[Start Communications Test]	Initiate communications test with PPads.
6	[Communications Test Status]	Initiate communications test results.
7	Disable Remote Alarm	Used to disable the remote alarm from activating. You then must go into the "Alarms" screen to enter the time you wish to disable the alarm for.

4-5 Configure Screens

4-5-1 Parameters Screen



PARAMETERS SCREEN
ILLUSTRATION 4-16

TABLE 4-14
PARAMETERS SCREEN

No.	Column	Description
1	Item Name	Name of the monitored parameter, and lint to the parameter setup.
2	En(abled)	Enable indicator. The display shows Y when parameter monitoring is enabled and N when disabled.
3	Units	Units of measure.
4	High Limit	Value above which causes an alarm condition.
5	Low Limit	Value below which causes an alarm condition.
6	Precision	Displayed precision of data (X.X, X.XX...).
7	Channel	Analog channel assignment (A/D Conv).
8	Coef Table	Coefficient Table. Indicates which coefficient table is used for error compensation.
9	Sample Count	Number of samples to record before averaging.

You can edit each value for each parameter by clicking on the Item Name. This opens a single parameter screen with edit fields for each value for the selected parameter. An example of a single parameter screen is shown below.

SHem Configure - Netscape

File Edit View Go Window Help

Back Forward Reload Home Search Guide Print Security Stop

Bookmarks Location: http://10.0.0.5/masco.html

Status Diagnose Configure

Parameters

1 [Back] 2 [Check] 3 [Save]

Name	Current Value	New Value
4	Y 5	Yes: ☒ No: ☐ 6
Item Name	Coldhead Temp	Coldhead Temp
Units	K	K
High Limit	6	6
Low Limit	3	3
Pre-cision	2	2
Technique	RuO Diode	RuO Diode
Channel	7	7
Coef Table	12	12
Sample Count	24	24
Note 1	Can enter text here if needed to explain situation	
Note 2		
Note 3		
Note 4		

TECHNIQUE, CHANNEL, COEF TABLE AND SAMPLE COUNT SHOULD NOT BE CHANGED.

Document: Done

SINGLE PARAMETER SCREEN
ILLUSTRATION 4-17

TABLE 4-15
SINGLE PARAMETER SCREEN

No.	Name	Description
1	[Back] button	Returns you to the Configure Parameters screen.
2	[Check] button	
3	[Save] button	Saves the new parameter values.
4	Name field	Name of the parameter whose values are being modified.
5	Current Value field	Displays the current parameter value settings.
6	New Value field/toggle	Fields and toggle for typing in new values or toggling to a different setting.

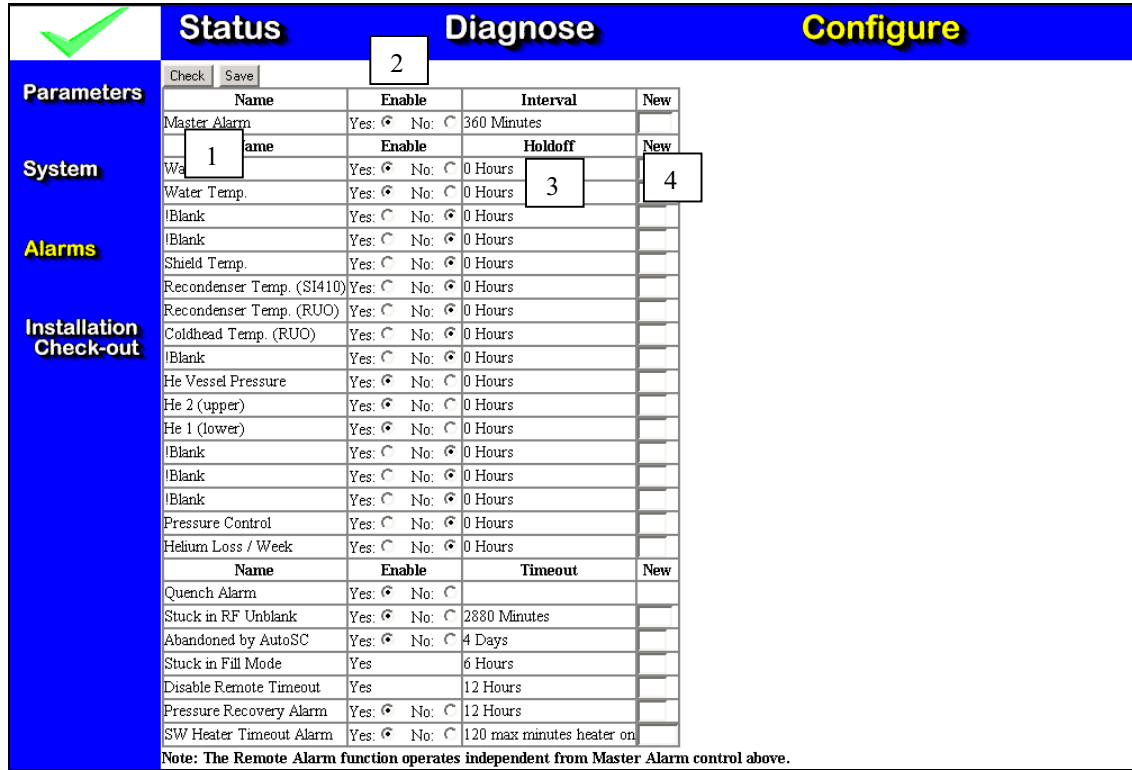
4-5- 2 System Screen

SYSTEM SCREEN
ILLUSTRATION 4-18

TABLE 4-16
SYSTEM SCREEN

No.	Name	Description
1	System Information	Reflects data entered during the install process.

4-5-3 Alarms Screen



Parameters		Name	Enable	Interval	New
System	Master Alarm	Yes: ☒ No: ☐	360 Minutes		
	Water Temp.	Yes: ☒ No: ☐	0 Hours		
	Water Temp.	Yes: ☒ No: ☐	0 Hours		
Alarms	Blank	Yes: ☐ No: ☒	0 Hours		
	Blank	Yes: ☐ No: ☒	0 Hours		
	Shield Temp.	Yes: ☐ No: ☒	0 Hours		
Installation Check-out	Recondenser Temp. (SI410)	Yes: ☐ No: ☒	0 Hours		
	Recondenser Temp. (RUO)	Yes: ☐ No: ☒	0 Hours		
	Coldhead Temp. (RUO)	Yes: ☐ No: ☒	0 Hours		
	Blank	Yes: ☐ No: ☒	0 Hours		
	He Vessel Pressure	Yes: ☒ No: ☐	0 Hours		
	He 2 (upper)	Yes: ☒ No: ☐	0 Hours		
	He 1 (lower)	Yes: ☒ No: ☐	0 Hours		
	Blank	Yes: ☐ No: ☒	0 Hours		
	Blank	Yes: ☐ No: ☒	0 Hours		
	Blank	Yes: ☐ No: ☒	0 Hours		
	Pressure Control	Yes: ☐ No: ☒	0 Hours		
	Helium Loss / Week	Yes: ☐ No: ☒	0 Hours		
	Quench Alarm	Yes: ☒ No: ☐			
	Stuck in RF Unblank	Yes: ☒ No: ☐	2880 Minutes		
	Abandoned by AutoSC	Yes: ☒ No: ☐	4 Days		
	Stuck in Fill Mode	Yes	6 Hours		
	Disable Remote Timeout	Yes	12 Hours		
	Pressure Recovery Alarm	Yes: ☒ No: ☐	12 Hours		
	SW Heater Timeout Alarm	Yes: ☒ No: ☐	120 max minutes heater on		
		Name	Enable	Timeout	New

Note: The Remote Alarm function operates independent from Master Alarm control above.

Alarms Screen

ILLUSTRATION 4-19

TABLE 4-17
ALARMS SCREEN

No.	Name	Description
1	Name field	Name of the alarm parameter.
2	Enable	Enables or disables the alarm for the selected parameter.
3	Interval/Holdoff	Interval: time for holding off master alarm in minute values between 1 and 720 minutes(12 hours). Holdoff: Set time to delay before enabling an alarm in hours between 1 and 99. Clicking on the <i>Check</i> button will display the time left for alarm hold-off.
4	New	Allows new values for Interval and Holdoff to be entered.

4-5-4 Installation Check-out Screen

Site_Server Configuration Settings	
Product Type	HCS Site Server
Site Server or Hospital Name	Test bAY Bay 4
Site Server Location	GEMR
Site Server ID	000SHEMGE4
Site Server IP Address	172.28.17.1/255.255.255.0
Additional IP Addresses	172.28.12.1/255.255.255.0
Modem Type	Internal_US_Robotics_288
Site Server Phone Number	8w4145216874
Site Server Link ID	342471
Site Server Chat Secret	bdabchee936
Center Portmaster Name	olc-pm1
Center Portmaster Address	3.87.88.221
GE IP addresses and subnets	3.245.68.0/255.255.252.0 3.57.88.0/255.255.252.0 3.87.168.0/255.255.252.0 3.87.88.0/255.255.252.0 3.70.152.0/255.255.252.0 3.245.32.0/255.255.252.0 3.45.16.0/255.255.252.0 3.245.60.0/255.255.252.0 3.245.250.0/255.255.252.0 3.7.192.0/255.255.252.0 3.245.172.0/255.255.252.0 3.87.16.0/255.255.252.0 3.7.188.0/255.255.252.0

INSTALLATION CHECK-OUT SCREEN
ILLUSTRATION 4-20

This screen reflects data downloaded to the MM by the OLC after initial check-out.

5- SYSTEM CONFIGURATION

SHeM Configure - Microsoft Internet Explorer provided by GE Medical Systems

File Edit View Favorites Tools Help

Back Forward Stop Home Search Favorites History Print View Source

Address <http://3.45.82.67/masco.html> Go

Status Diagnose Configure

Parameters

Save

System Type	Lightning
Magnet Type	LCC
Shield Cooler Type	Sumitomo
Magnet Serial Number	007
MAC Team Leader	
Primary FE	

Done Internet

SYSTEM CONFIGURATION SCREEN
ILLUSTRATION 5-1

Go to the Configure System screen.

The information displayed here is from the original setup script, and is not editable.

You can change the following:

- Magnet type
- Magnet Serial number
- MAC team leader
- Primary FE

6- PARAMETER SETUP

6-1 Regular Parameter Set-up

Item Name	En	Units	High Limit	Low Limit	Precision	Technique	Channel	Coef Table	Sample Count
Water Flow	Y	GPM	4.5	1.5	1	AD Reading	0	1	6
Water Temp.	Y	Deg F	70	35	1	AD Reading	1	2	3
Blank	N		0	0	1	AD Reading	2	0	0
Blank	N		0	0	1	AD Reading	3	0	0
Shield Temperature	Y	K	50	30	2	AD Reading	4	11	4
Recondenser (SI-410) Temp	Y	K	6	3	2	AD Reading	5	11	4
Recondenser (RUO) Temp	Y	K	30	3	2	RuO Diode	6	12	4
Coldhead (RUO) Temp	Y	K	30	3	2	RuO Diode	7	12	4
Blank	N		0	0	1	AD Reading	8	0	0
He Vessel Pressure	Y	PSIG	6.25	1	2	AD Reading	9	4	16
He 2 (upper)	Y	%	88	68	1	He Sensor	10	10	4
He 1 (lower)	Y	%	99.9	80	1	He Sensor	11	10	4
Blank	N		0	0	1	AD Reading	12	0	0
Blank	N		0	0	1	AD Reading	13	0	0
Blank	N		0	0	1	None	14	0	1
Pressure Control	Y	PSIG	1.6	1.3	2	PSI Control	9	0	1
Helium Loss / Week	N	%/Wk	5	0.5	2	None	11	N.A.	N.A.

CONFIGURE PARAMETERS SCREEN
ILLUSTRATION 6-1

1. Go to the Configure Parameters screen.
2. Click on an Item Name to edit the fields for that parameter. See table 6-1 for default values for each parameter for the different magnet types.
3. While it is possible, using this method, to alter various parameter settings, it is recommended that you do not change these settings unless specifically required to do so.

IMPORTANT

The enable value should always be Y for monitored parameters. Use the Alarm's page to turn off or disable individual alarms.

TABLE 6-1
PARAMETER VALUES

HFO magnets				
Parameter name	High Limit	Low Limit	Precision	Enabled
Water flow	4.5	1.5	1	Y
Water temp	85	40	1	Y
Shield temp (single stage)	35	20	2	N
Shield temp (dual stage)	50	25	2	N
Recondenser (SI-410)	6	3	2	N
Recondenser (RUO)	6	3.9	2	N
Magnet vessel pressure	6.25psig	0.8psig	2	Y
Helium level (upper)	110%	75% See note 1	1	Y
Helium level (lower)	110%	95% See Note 1	1	Y
Pressure control	0.9	1.0	1	Y

Note 1 For HFO Magnets set the helium low level of the upper vessel to 75% at the time of magnet monitor checkout, or after ramp/fill. The magnet monitor would then alarm if the magnet would develop a slow leak. If the helium level for the upper vessel were to drop below 75%, the lower level must be adjusted to a value that is 5 to 10% below the actual level. The recommended fill level for the HFO magnet is 75% of total magnet vessel, or no less than 50% on the upper vessel. The low limit for the upper vessel should not go below 50%. The low limit for the lower vessel should not go below 85%.

6-2 Coldhead Cycling

The HFO system must turn off the two-stage cryo-cooler coldhead during imaging for vibration issues. The Magnet Monitor sends data to the host via the Ethernet cable, run 942. The following conditions will disable coldhead cycling (keeping the two-stage cryo-cooler on during scanning):

- Vessel pressure is out of the alarm limits set by magnet monitor. Limits for vessel pressure shall not exceed 6.25 psig.
- Helium level for either vessel is out of the alarm limits set by magnet monitor. See Note 1 above.
- There is no communication between the host and magnet monitor

Therefore it is important that the parameter limits are set per table 6-1.

Note

The Host system will only reference the alarm limits, which are set in the Configure/Parameters section of magnet monitor. The Host will ignore alarm hold-off for individual alarms or the Master alarm.

Individual Parameter Screen

Name	Current Value	New Value
En	Y	Yes: ☒ No: ☐
Item Name	Coldhead Temp	Coldhead Temp
Units	K	K
High Limit	6	6
Low Limit	3	3
Pre-cision	2	2
Technique	RuO Diode	RuO Diode
Channel	7	7
Coef Table	12	12
Sample Count	24	24
Note 1		
Note 2		
Note 3		
Note 4		

INDIVIDUAL PARAMETER SCREEN
ILLUSTRATION 6-2

This is the screen you would see if you clicked on an Item Name from the configure Parameters screen(Illustration 6-1).

These parameters can be altered by typing a new value into a field and pressing [Save]. It is not recommended that these selections be modified unless specifically required to do so.

IMPORTANT

The enable value should always be Y for monitored parameters. Use the Alarm's page to turn off or disable individual alarms.

7- ALARM SETUP

7-1 Regular Alarm Setup

	Status	Diagnose	Configure																																																
	Check Save																																																		
	Parameters	<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 30%;">Name</th> <th style="width: 20%;">Enable</th> <th style="width: 30%;">Interval</th> <th style="width: 20%;">New</th> </tr> </thead> <tbody> <tr> <td>Master Alarm</td> <td>Yes: ☒ No: ☐</td> <td>360 Minutes</td> <td> </td> </tr> </tbody> </table>			Name	Enable	Interval	New	Master Alarm	Yes: ☒ No: ☐	360 Minutes																																								
Name	Enable	Interval	New																																																
Master Alarm	Yes: ☒ No: ☐	360 Minutes																																																	
System	<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 30%;">Name</th> <th style="width: 20%;">Enable</th> <th style="width: 30%;">Holdoff</th> <th style="width: 20%;">New</th> </tr> </thead> <tbody> <tr> <td>Water Flow</td> <td>Yes: ☒ No: ☐</td> <td>0 Hours</td> <td> </td> </tr> <tr> <td>Water Temp.</td> <td>Yes: ☒ No: ☐</td> <td>0 Hours</td> <td> </td> </tr> </tbody> </table>			Name	Enable	Holdoff	New	Water Flow	Yes: ☒ No: ☐	0 Hours		Water Temp.	Yes: ☒ No: ☐	0 Hours																																					
Name	Enable	Holdoff	New																																																
Water Flow	Yes: ☒ No: ☐	0 Hours																																																	
Water Temp.	Yes: ☒ No: ☐	0 Hours																																																	
Alarms	<table border="1" style="width: 100%; border-collapse: collapse;"> <tbody> <tr> <td>Blank</td> <td>Yes: ☐ No: ☒</td> <td>0 Hours</td> <td> </td> </tr> <tr> <td>Blank</td> <td>Yes: ☐ No: ☒</td> <td>0 Hours</td> <td> </td> </tr> <tr> <td>Shield Temp.</td> <td>Yes: ☐ No: ☒</td> <td>0 Hours</td> <td> </td> </tr> </tbody> </table>			Blank	Yes: ☐ No: ☒	0 Hours		Blank	Yes: ☐ No: ☒	0 Hours		Shield Temp.	Yes: ☐ No: ☒	0 Hours																																					
Blank	Yes: ☐ No: ☒	0 Hours																																																	
Blank	Yes: ☐ No: ☒	0 Hours																																																	
Shield Temp.	Yes: ☐ No: ☒	0 Hours																																																	
Installation Check-out	<table border="1" style="width: 100%; border-collapse: collapse;"> <tbody> <tr> <td>Recondenser Temp. (SI410)</td> <td>Yes: ☐ No: ☒</td> <td>0 Hours</td> <td> </td> </tr> <tr> <td>Recondenser Temp. (RUO)</td> <td>Yes: ☐ No: ☒</td> <td>0 Hours</td> <td> </td> </tr> <tr> <td>Coldhead Temp. (RUO)</td> <td>Yes: ☐ No: ☒</td> <td>0 Hours</td> <td> </td> </tr> <tr> <td>Blank</td> <td>Yes: ☐ No: ☒</td> <td>0 Hours</td> <td> </td> </tr> <tr> <td>He Vessel Pressure</td> <td>Yes: ☒ No: ☐</td> <td>0 Hours</td> <td> </td> </tr> <tr> <td>He 2 (upper)</td> <td>Yes: ☒ No: ☐</td> <td>0 Hours</td> <td> </td> </tr> <tr> <td>He 1 (lower)</td> <td>Yes: ☒ No: ☐</td> <td>0 Hours</td> <td> </td> </tr> <tr> <td>Blank</td> <td>Yes: ☐ No: ☒</td> <td>0 Hours</td> <td> </td> </tr> <tr> <td>Blank</td> <td>Yes: ☐ No: ☒</td> <td>0 Hours</td> <td> </td> </tr> <tr> <td>Blank</td> <td>Yes: ☐ No: ☒</td> <td>0 Hours</td> <td> </td> </tr> <tr> <td>Pressure Control</td> <td>Yes: ☐ No: ☒</td> <td>0 Hours</td> <td> </td> </tr> <tr> <td>Helium Loss / Week</td> <td>Yes: ☐ No: ☒</td> <td>0 Hours</td> <td> </td> </tr> </tbody> </table>			Recondenser Temp. (SI410)	Yes: ☐ No: ☒	0 Hours		Recondenser Temp. (RUO)	Yes: ☐ No: ☒	0 Hours		Coldhead Temp. (RUO)	Yes: ☐ No: ☒	0 Hours		Blank	Yes: ☐ No: ☒	0 Hours		He Vessel Pressure	Yes: ☒ No: ☐	0 Hours		He 2 (upper)	Yes: ☒ No: ☐	0 Hours		He 1 (lower)	Yes: ☒ No: ☐	0 Hours		Blank	Yes: ☐ No: ☒	0 Hours		Blank	Yes: ☐ No: ☒	0 Hours		Blank	Yes: ☐ No: ☒	0 Hours		Pressure Control	Yes: ☐ No: ☒	0 Hours		Helium Loss / Week	Yes: ☐ No: ☒	0 Hours	
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Note: The Remote Alarm function operates independent from Master Alarm control above.

CONFIGURE ALARMS SCREEN

ILLUSTRATION 7-1

1. Go to the Configure Alarms screen.
2. Set the following parameters to the specified values:
 - Enable (Yes/No)
Normally all parameters that are being monitored must be enabled.
 - New Interval/Holdoff
Interval is the amount of time between MM callouts (to OLC) if alarms exist.
Holdoff is a temporary countdown timer that will prevent a particular alarm condition from being reported.
3. Click on the [Save] button.
4. Helium Loss/Week is not currently functional.

7-2 New alarms for HFO

There are several new alarms for the HFO magnet. See Table 7-1 for details.

Alarm	Type	Details
Pressure Recovery		The magnet monitor will evaluate how fast the vessel pressure recovers after customer is done scanning for the day. A slow recovery indicates poor thermal performance. The Magnet Monitor will alarm if pressure recovery is less than 0.5psi over a 2 hour period after scanning is complete.
SW Heater timeout		If the magnet vessel pressure is low, and is keeping the magnet heater on for an extended period, the magnet monitor will alarm and turn the heater off.
UPS Failure		Indicates that there has been a power failure and the UPS will shut down.
Helium meter error		There is a device that monitors the 75ma current output to the helium meter. If this current is out of tolerance the helium readings may not be valid.
Latched Pressure		Indicates that the vessel pressure value has not changed for 10 minutes. The magnet monitor would then reset. If the vessel pressure remains constant for an additional 10 minutes the magnet monitor would alarm to OLC.
Non-connect		The HFO magnet Monitor is configured to call the PPADS system on a daily basis. If a Magnet Monitor does not connect with PPADS, a dispatch will be created at the specific OLC.

7-3 Remote Alarm Activation

The remote alarm will activate under the following conditions:

- Helium level for either vessel is out of limits
- RUO temperatures are out of limits for more than 90 minutes
- Magnet Monitor software is not functioning properly

7-4 Alarm Activation

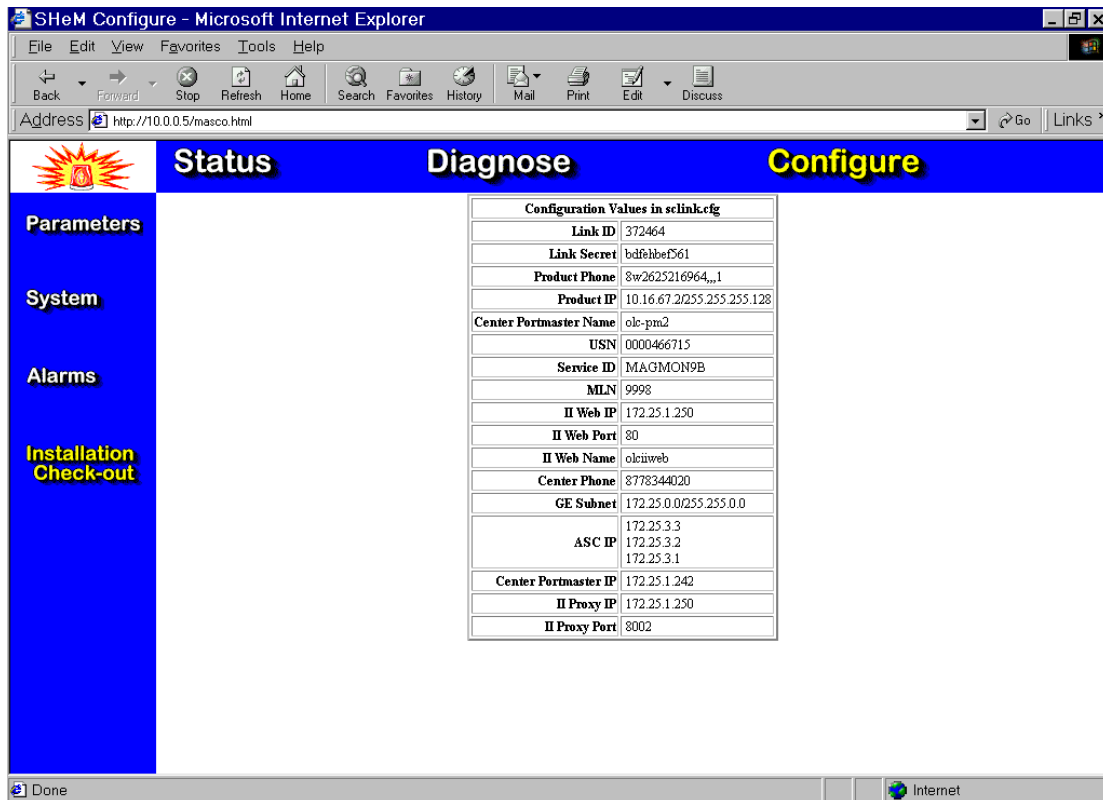
All alarms noted in Illustration 7-1 can be Enabled (select **Yes**) or Disabled (select **No**).

The “digital” alarms listed in Table 7-1 will report to the OLC.

TABLE 7-1
ALARM STATUS

Alarm	Alarm to OLC?	Can Disable	Explanation
Quench Alarm	Y	Y	Activates when magnetic reed switch does not detect magnetic field
Stuck in RF Unblank	Y	Y	Activates when RF unblank signal is activated for greater than 6 hours and System cabinet power is detected
Abandoned by AutoSC	Y	Y	Activates when connection from AutoSC to magnet monitor has failed. Mag Mon can possibly still dial out to AutoSC
Stuck in Fill mode	Y	Y	Activates when key is in “Fill” position has been on for greater than number of minutes in the Timeout section.
Disable remote timeout	N	N	Used for HFO only
SW heater timeout	Y	Y	Activates when magnet heater has been on for greater than number of minutes in the Timeout section. Value for LCC magnet should be 5000.
Unable to reset compressor	Y	N	Automatic restart of compressor failed. Compressor restart setup is part of magnet monitor configuration setup.
Fill Notification	Y	N	Sends fill data to OLC
Latched pressure detected	Y	N	Activates if magnet monitor reports the same pressure reading for 60, 1 minute samples.
UPS Power Failure	Y	N	Activates when magnet monitor is on UPS power.
Helium meter error	Y	N	Activates when magnet monitor does not detect proper current going out to magnet.
Touch	Y	N	Used for verifying remote connection from magnet monitor to PPADS.

8- System Functional Tests



CONFIGURE INSTALLATION CHECK-OUT SCREEN
ILLUSTRATION 8-1

You can view the following parameters:

- Link ID
- Link Secret
- Product phone
- Product IP
- Center Portmaster name
- USN
- Service ID
- MLN
- II Web IP
- II Web Port
- II Web Name
- Center Phone
- GE Subnet
- ASC IP
- Center Portmaster IP
- II Proxy IP & Port

3. These parameters are set by the OLC after check-out is successfully performed.

9- TROUBLESHOOTING

9-1 Troubleshooting tables

TABLE 9-1
MAGNET MONITOR FAULT ISOLATION GUIDELINES

Symptom	Suggested Actions
Your local support center is unable to call into the Magnet Monitor	Check phone connections between the Magnet Monitor and the customer phone line. Try bypassing the multiplexer box. Make sure the support center has the correct phone number for this site, including the multiplexer code if applicable. Run the COMMUNICATION TEST command and make sure the Magnet Monitor can dial out. The test can be found in the Diagnostic/Remote field. Verify the site phone line connection by plugging in a telephone. Try calling this telephone from outside the site. If using an external modem, verify that it is configured to auto-answer incoming calls.
The Magnet Monitor does not call PPADS when an alarm is detected.	Verify that the Master Alarm time has elapsed, alarms are not reported immediately. Go to the Alarms screen and verify that the minute for master alarm is zero. Make sure the dial-out prefix on the magnet monitor is correctly set for the phone system in the MR suite. Example is we may need a 9,1 rather than a 9 or a 1 rather than no prefix. Verify that the parameter in question does not have a hold-off time set, check in Configure/Alarms Verify that the MASTER ALARM in the status/alarm field is set to enable. Also verify that all pertinent alarms are enabled. Check the COMMS field to see if there is any error messages that might assist you in determining the problem. The COMMS field is found in Status/Log/Comms. Run the Toucholc.pl command and make sure the Magnet Monitor can dial out. The test can be found in the Status/Remote field. Verify the phone line can make outgoing calls by plugging a telephone into the line and calling PPADS, or a different 800 number. The PPADS server may be inoperative, try again later. The phone system at the site may be trying to block toll free numbers. Work with the customer to change the phone system programming.
The Magnet Monitor does not properly reset.	Unless specifically requested to boot from a floppy disk, verify that there is no floppy disk in the drive.
Spike noise in the image	Check all connectors for tight mechanical connections, jackscrews and other hardware must be secure. Verify that none of the connectors are laying against the magnet, or other conductive surfaces.

TABLE 9-2
PARAMETER FAULT ISOLATION

Symptom	Suggested Actions
Message ER1 appears on the display	This means that there is an open circuit in the helium cable between the Magnet Monitor and the magnet, or the Magnet Monitor is not sending the proper current to the magnet for an accurate helium reading. Check cables between magnet and Magnet Monitor. If continuity is OK check for 75ma coming out of the magnet monitor
Message ER2 appears on the seven segment display	This means that the magnet heater is stuck on, possibly due to a software hang or other circumstance. Latched pressure was detected and a reboot of the magnet monitor has already occurred. If problem occurs again the Magnet Monitor must be replaced. Latched pressure indicates that the PC in the magnet monitor could have a software hang.
Message ER3 appears on the seven segment display	Heater timeout. The magnet heater was being held on for an extended period. Possible cause might be a slow leak in the magnet, and vessel pressure would never reach the upper limit. In this case the heater would turn off to conserve helium.
Shield temperature/s out of range.	If temperature reading is either 0K or 592K, check cabling between Magnet Monitor and sensors. This type of reading can indicate an open circuit. Check the shield temperature/s using a lakeshore thermometer. If the measured temperatures differ check the continuity of the cables between the Magnet Monitor and the sensors. Verify the 10 ua current source by placing a microamp meter in series between pins 1 & 2 or 11 & 12 or 15 & 16 or 22 & 23, of the J8 connector at the Magnet Monitor. The first pin is positive, the second is negative. Disconnect the cable while performing test.
RuO temps out of range	If temperature reading is either 1.5K or 24K, check cabling between Magnet Monitor and sensors. This type of reading can indicate an open circuit. Also verify that the RuO Buffer amplifier, located at the magnet, is properly connected. Check the RuO temperature/s using an RUO meter. If the measured temperatures differ check the continuity of the cables between the Magnet Monitor and the sensors. Verify the 10 ua current source by placing a microamp meter in series between pins 1 & 2 or 11 & 12 or 15 & 16 or 22 & 23, of the J8 connector at the Magnet Monitor. The first pin is positive, the second is negative. Disconnect the cable while performing test.
Vessel pressure out of range	If the vessel pressure reading is close to zero, check the actual pressure in the magnet using the mechanical gauge. The cabling between the pressure transducer and the Magnet Monitor can be checked. Disconnect the cable at the pressure transducer. Place a 1.5vdc battery on pin 7 positive side and pin 6 negative side of J8 (Scan Room) at the Magnet Monitor. The Magnet Monitor should then read about 1.8 psi.

TABLE 9-2 (Continued)
PARAMETER FAULT ISOLATION

Symptom	Suggested Actions
Helium level out of range	Verify ER1 does not appear during cycling on the magnet monitor LED display. If you suspect there is a problem with the helium reading, perform helium meter calibration, located in the "Magnet Monitor Hardware Installation" Section 6. If the helium reading is consistently around 100% check the cabling between the Magnet Monitor and the helium sensors in the magnet. Verify cables going from the magnet monitor to the magnet are not swapped. The only location where the cables could be swapped is at the magnet interface box.
Compressor Water flow or water temperature out of range.	Check water source to compressor. Check connections to water flow meter. Verify that water is flowing. Check for condensation in the flow/temp box.
RF-Unblank always active or inactive	If the RF-Unblank is always active, a logic low at pin 6 (+) and pin 7 (-) of the cable coming into J10 on the Magnet Monitor, check cabling at system cabinet. The cable used for the Unblank signal should connect to J9 (RF UNBLNK 1) of the IPG board in the Systems cabinet. Check continuity of the cabling between the UNBLANK connection in the systems cabinet and the Magnet Monitor.
System Power always active or inactive	Verify that the "Line trigger" signal from the PDU is correct. This is an AC 60Hz 6.3 Vrms signal. This signal should be seen at the connector coming into J10 of the Magnet Monitor, on pins 10 (+) and 11 whenever the system cabinet is powered up.
No Field is active.	Verify the reed switch is functioning. Disconnect the wires at the reed switch and short them together. If the NO FIELD message appears at the Magnet Monitor, the cabling back to the Magnet Monitor is functioning. An "open" circuit indicates FIELD (digital 0), a "short" indicates NO FIELD (digital 1). Change the orientation of the reed switch while in the magnetic field to try and activate the reed switch properly.
"Low Helium Level" message appears at the console	Jumper pins 4 and 6 together on connector J5(cryo mon) on the back of the TYME board in the SC cabinet.
Modem will not answer call from OLC, but Magnet Monitor can dial out.	The internal modem will not respond to a digital phone system. You can try putting a Mux box in series between the phone line and the Magnet Monitor. Check with hospital to verify phone line will accept incoming calls. Check for correct phone number & area code on PPADS database @ OLC.
Magnet Monitor cannot communicate with the Host via the Ethernet cable	Verify both the magnet monitor and the Host can communicate by trying to "ping" the host from the Magnet Monitor, and also Magnet Monitor to the Host. Magnet Monitor must have Insite Checkout

	• Verify the Host has the correct IP addressing. Open a C-shell on the host. Type cd /etc. Type more hosts Verify IP information for host and Magnet Monitor are correct. • Verify the Magnet Monitor has correct IP addressing. Connect your laptop to the Magnet Monitor. Login using Hyperterminal. Login name is root, Password is mvserve. • To verify correct Magnet Monitor information: Type cd /etc/sysconfig/network-scripts Type more ifcfg-eth0 The following information should be displayed: DEVICE=eth0 BOOTPROTO=static IPADDR=3.45.82.82 (IP address of Magnet Monitor) NETMASK=255.155.252.0 (netmask of Magnet Monitor) ONBOOT=yes This verifies that the Magnet Monitor has the correct IP addressing for the Magnet Monitor. Type cd ~syscon Type more .rhosts 26.45.82.80(IP address of Host) This verifies that the Magnet Monitor has the correct IP addressing for the host.
--	---

IMPORTANT

If you need to reset, or power down the Magnet Monitor, do the following steps to properly shut down the software.

- Hold the *Sample* button in for about four seconds. **halt** should appear on the display.
- Release the *Sample* button for less than 1 second, and then press the *Sample* button again. **oooo** (four lower case o's) should appear on the display.
- Wait for at least **90 seconds**, then you may reset or power down the Magnet Monitor.

9-2 Boot-up Timeline

The following will describe the sequential events that will occur as a magnet monitor is re-booted or powered up. The times noted are approximate.

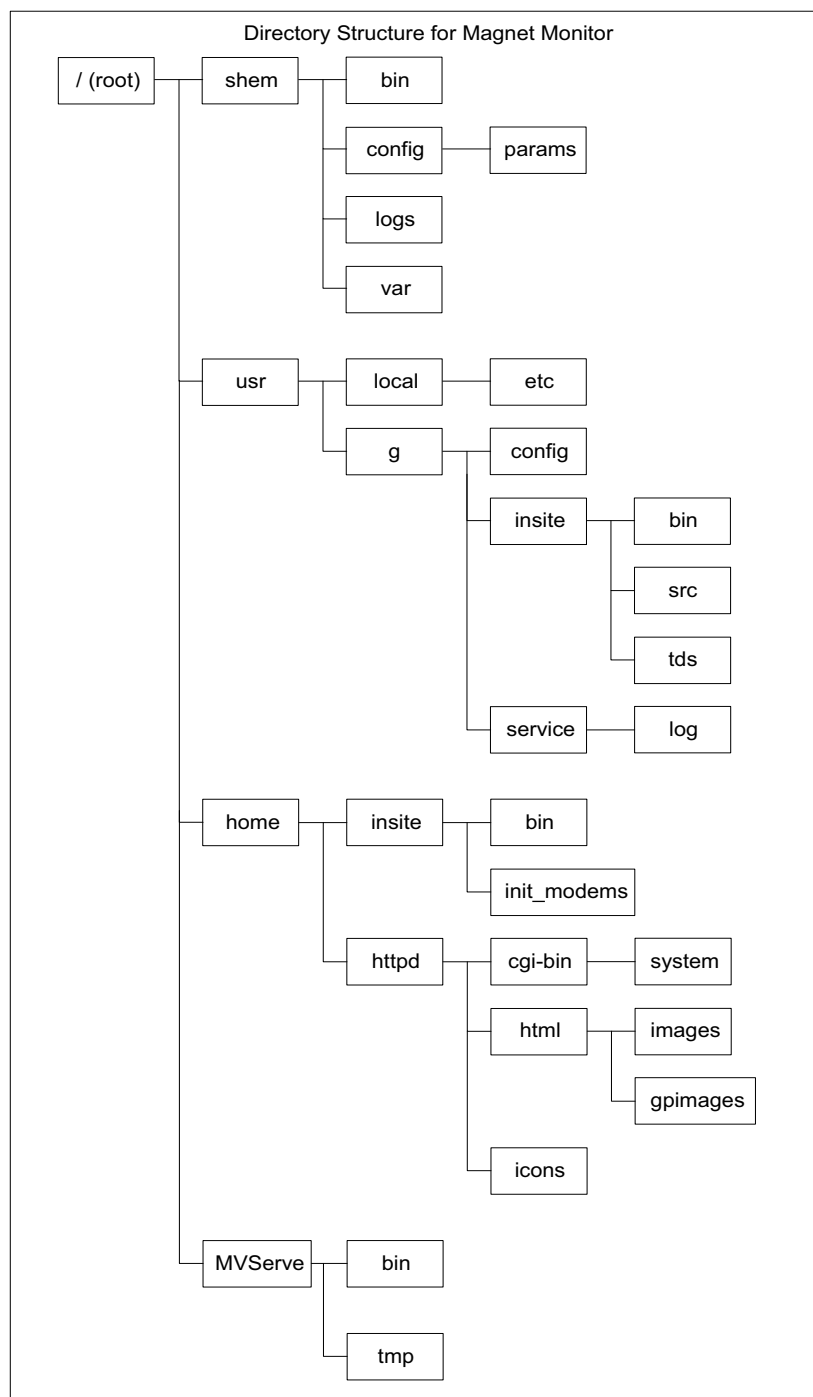
- 10 sec The HDD light located on the left side of the unit will come on as the hard drive is accessed. It will flicker during the remainder of the boot-up cycle
- 45 sec The display will come on with **8.8.8.8**. displayed, all LED's will luminate
- 60 sec The helium reading will activate and display the helium level in the magnet and cycle between levels and vessel pressure will begin.

Note

Do not reset or power down the magnet monitor during the boot-up process. Software may be corrupted during this time.

9-3 Software Structure

Four new branches have been added to the standard Linux directory tree. The shem branch contains all of the files for the Magnet Monitor pertaining to reading parameter data, logging the data, reporting alarms, etc. The /usr/g branch contains all of the files used by the IP/TDS system that is used to report alarm conditions to the AutoSC. These files are also used during remote access of the Magnet Monitor from the Online Center. The /home/httpd branch contains the files used by the Web Server. The /MVserve branch contains the InitConfig and ConfigLink scripts that are used to do the Insite Checkout procedure. See Illustration 9-2.



SOFTWARE DIRECTORY STRUCTURE
ILLUSTRATION 9-2

9-4 Files on Proprietary Disk

There are 7 files on the proprietary floppy.

TABLE 9-2
PROPRIETARY SOFTWARE FILES

Filename	Size	Description
Installn.tgz	67K	Compressed tar non-proprietary software
Installn	2K	Executes non-proprietary install over proprietary mode
Installp	1K	Executes proprietary software install
Installp.tgz	332K	Compressed tar proprietary software
Mdmcbx.inf	4K	Null modem file needed for laptop communication

9-5 Dumping Magnet Monitor Data to a Floppy

If needed, you can store the log files from a magnet monitor onto a floppy using the following steps. The data files are in binary format and cannot be viewed. You would have to transfer the files to another magnet monitor in order to view the files.

1. Connect a laptop to the magnet monitor using a 9 pin female to 9 pin female null modem cable.
2. Open a hyperterminal window. See section 2 of this manual for the details.
3. At login prompt type **root**, password is **mvserve**.
4. Install a blank floppy into the Magnet Monitor and mount the floppy:

mount -t msdos /dev/fd0 /mnt/floppy

5. Copy the files: **cp /shem/logs/* /mnt/floppy**
6. Un-mount the floppy: **umount /dev/fd0**
7. To load the copied files on to a different Magnet Monitor connect a laptop to the magnet monitor using a 9 pin female to 9 pin female null modem cable.
8. Open a hyperterminal window. See section 2 of this manual for the details.
9. At login prompt type **root**, password is **mvserve**.
10. Install the floppy containing the data into the magnet monitor and mount the floppy:

mount -t msdos /dev/fd0 /mnt/floppy

11. Copy the files: **cp /mnt/floppy/* /shem/logs**
12. Un-mount the floppy: **umount /dev/fd0**

9-6 Changing time and date

If needed you may change the time and date of the magnet monitor using the following steps.

1. Connect a laptop to the magnet monitor using a 9 pin female to 9 pin female null modem cable.
2. Open a hyperterminal window. See section 2 of this manual for the details.
3. At login prompt type **root**, password is **mvserve**.
4. At the prompt type in **date MMDDhhmmyy** where
MM is the 2 digit month
DD is the 2 digit day
hh is the 2 digit hour in military time
mm is the 2 digit minute
yy is the 2 digit year

Example is **date 0816132001** for August 16, 2001 at 1:20pm.

5. To write the date into firmware type **/sbin/clock --systohc** If you omit this step the magnet monitor will loose the time and date when rebooted.
6. Type **reboot** to reset the Magnet Monitor.

IMPORTANT

If you need to reset, or power down the Magnet Monitor, do the following steps to properly shut down the software.

- Hold the *Sample* button in for about four seconds. **halt** should appear on the display.
- Release the *Sample* button for less than 1 second, and then press the *Sample* button again. **oooo** (four lower case o's) should appear on the display.
- Wait for at least **90 seconds**, then you may reset or power down the Magnet Monitor.

9-7 Determining Software Revision for the Magnet Monitor

There are two methods to determine the software revision on a Magnet Monitor.

1. The software revision will be displayed when connected through the web browser on the "Status/System" page, or the "Configure/Installation Check-Out" page.
2. Use Hyperterminal to determine the software revision of the magnet monitor. Log into the Magnet Monitor.

When the login prompt appears the software revision is displayed:

Magnet Monitor Proprietary Software, Rev 2.0

Red Hat Linux release 6.1 (Cartman) Kernel Rev 2.2.12-20

The same information can be viewed by logging in to the Magnet Monitor, Login name is **root**, password is **mvserve**.

At the prompt type in **getver**. The same revision information should be displayed.

9-8 Archiving Insite Information

You can archive checkout information to a floppy. In the event that a magnet monitor is replaced, the checkout information can be restored. This step eliminates the need for an Insite Checkout after magnet monitor replacement.

1. Insert a blank floppy into the magnet monitor floppy drive.
2. Open a C-shell on the console. Telnet to the Magnet Monitor by typing **telnet x.xx.xx.xx** where x.xx.xx.xx is the IP address of the magnet monitor.

Login name is **root** (case sensitive)
Password is **mvserve** (case sensitive)

Note

You can also connect your laptop to the Magnet Monitor and use Hyperterminal to communicate. See Section 2 for proper settings.

3. Mount the floppy by typing **mount -t msdos /dev/fd0 /mnt/floppy**
4. Copy the files by typing
cp /usr/g/insite/sclink.cfg /mnt/floppy
cp /shem/config/alarm.cfg /mnt/floppy
cp /shem/config/param.cfg /mnt/floppy
cp /etc/sysconfig/network-scripts/ifcfg-eth0 /mnt/floppy
cp /home/syscon/.rhosts /mnt/floppy/rhosts (must be exact)
cp /shem/logs/.shemtype /mnt/floppy/shemtype (must be exact)
5. Unmount the floppy by typing **umount /dev/fd0**
6. Remove floppy from the drive, label as **Insite Information** and store in a safe location.

10- RETURNING TO NON-PROPRIETARY CONFIG

The following process describes how to install Non-Proprietary software onto a Magnet Monitor. The process assumes that your Magnet Monitor is operating.

After completing this procedure inform the On-Line Center in your pole that the Magnet Monitor software was set to Non-Proprietary mode so the site can be removed from the PPADS database.



**VERIFY THAT YOU HAVE THE PROPER PROPRIETARY DISKETTE INSTALLED.
FAILURE TO USE THE PROPER DISK WILL DESTROY THE HART DRIVE ON THE
MAGNET MONITOR.**

- ◆ Use disk 2252032 if you have the Granite Magnet Monitor, without the Pacemaker FMI installed.
- ◆ Use disk 2252032-3 if you have the Granite Magnet Monitor with the Pacemaker FMI installed.
- ◆ Use disk 2252032-3 if you have the Advantech Magnet Monitor.

IMPORTANT

Verify that the pressure control settings are properly set prior to returning to Non-Proprietary software. Heater should turn on when pressure drops to 0.9psig, and should turn off when pressure increases to 1.0psig.

1. Using a serial connection (9-pin female to 9-pin female null modem), and a laptop computer, open a hyperterminal window. Refer to Section 2, Proactive Software Load.
2. At the Login prompt, type in **root**
3. At the password prompt, type in **mvserve**
4. Insert the proper Proprietary software floppy disk into the floppy drive on the M M.
5. Mount the drive by typing the following:
mount -t msdos /dev/fd0 /mnt/floppy (case sensitive, don't ignore spaces)
6. Press [Enter].
7. Copy files to /tmp directory: **cp /mnt/floppy/* /tmp**
8. Press [Enter]
9. After the copy function is completed, unmount the floppy drive: **umount /dev/fd0**
10. Press [Enter]

11. Remove floppy disk from drive
- 12.** Execute the install script: **/tmp/installn**
13. Press [Enter].
14. When the install script completes, the magnet monitor will re-boot. This will cause a disconnect in your laptop connection.
15. Connect to the Magnet Monitor by pressing [Enter] on the laptop. Login name is **root** password is **mvserve**.
16. Remove the proprietary files by doing the following:

 cd /tmp

 rm install*

This will remove the proprietary files on the hard drive.

The Magnet Monitor will now work in the non-proprietary mode. All the necessary magnet configuration information will remain on the hard drive. No other steps are necessary.

11- RENEWAL PARTS

11-1 Magnet Monitor

There are two FRU's associated with the Magnet Monitor.

The entire unit is a FRU. The part number is 2219400.

The Global Modem Part number is 2245794.

11-2 Related Parts

Table 11-1 contains some of the unique parts that may be required for Magnet Monitor.

TABLE 11-1
UNIQUE PARTS LIST FOR MAGNET MONITOR

Description	Part Number	FRU code
4 line multiplexer with 115VAC	46-328475P1	1
8 line multiplexer with 115VAC	46-328475P2	1
Water flow and temp sensor	46-320835P1	1
Desiccant bag for water flow/temp sensor	2135979	1
Proprietary Software Floppy	2252032-3	1

12- MAINTAINING DATA FOR A PARTICULAR MAGNET

The Magnet Monitor will continue to store Minute data, Hour data, and Day data. The data will over-write based on the memory allocation which is:

2048 data points for Minute data.

1024 data points for Hour data.

1024 data points for Day data.

If a Magnet Monitor is replaced for any reason the data should be transferred from the original magnet monitor to the new magnet monitor if possible. See procedure in Section 9-5 of this manual.

If a Magnet is replaced the data on the Magnet Monitor should be deleted.

1. Connect a laptop to the magnet monitor using a 9 pin female to 9 pin female null modem cable.
2. Open a hyperterminal window. See section 2 of this manual for the details.
3. At login prompt type **root**, password is **mvserve**.
4. Do the following: **cd /shem/logs**
rm *.log*
rm *.bak
rm *.dat

IMPORTANT

If you need to reset, or power down the Magnet Monitor, do the following steps to properly shut down the software.

- Hold the *Sample* button in for about four seconds. **halt** should appear on the display.
- Release the *Sample* button for less than 1 second, and then press the *Sample* button again. **oooo** (four lower case o's) should appear on the display.
- Wait for at least **90 seconds**, then you may reset or power down the Magnet Monitor.

13- GLOSSARY

Cryogen

A coolant in the form of a liquefied gas such as helium or nitrogen used to cool the magnet to superconducting temperatures.

Gauss

Unit of measurement for the strength of a magnetic field.

Generic Back Plane

A circuit board that has only interconnection hardware (such as an ISA bus) and no electronic circuitry.

HFO

Acronym for High Field Open

Industry Standard Architecture (ISA) Bus

A series of common connection points used to connect ISA compatible boards.

Low cryogen consumption magnet.

Magnet Interface Board

Provides a connection between the Magnet Monitor and the magnet and associated support hardware and electronics such as compressors and cooling shields. The board also drives the display board and conditions analog signals before sending them to the analog to digital converter board (ADC).

Modem

A device used to allow computers to communicate over telephone lines.

Network Interface Card (NIC)

Provides circuitry for connecting a computer to a network.

OLC

GE OnLine Center

Penetration Panel

An connection panel used to isolate the magnet room's electrical cables from external cables.

Single Board Computer (SBC)

A single board that contains the main computing hardware such as the central processing unit (CPU), memory, controllers for the disk drives, etc. This board differs from computer "mother boards" in that it does not have a built-in bus for connecting other boards.

APPENDIX A - LAPTOP SERIAL CONNECTION SETUP

This connection, and description, is to be used when first installing a Magnet Monitor. The Serial connection can be used at other times, with the same or different Logins and Passwords. This information will be provided for each case that it is needed. This type of connection to the Magnet Monitor is generally intended to launch script files, copy new files onto the internal hard drive or to interact directly with the Linux operating system.

TABLE A-1
PARAMETERS MONITORED BY MAGNET MONITOR

Step	Description	Comments
1	Connect Laptop to the Magnet Monitor using a Null Modem serial cable.	Connect to an available COM port on the Laptop, and to the "Laptop" connection on the Magnet Monitor.
2	Use the Hyperterminal connection in the laptop to connect to the Magnet Monitor as explained in Hardware Installation Manual, Section 5 (2230681)	The Hyperterminal screen comes up with the Linux prompt for Login.
3	Give login name as setup and password as monitor .	A menu asking the Service Engineer to select the type of magnet used appears.
4	Select the appropriate option for your Magnet type. The default is an LCC type magnet.	A message showing the type of magnet appears and the machine logs out the user and Hyperterminal connection is lost. After a while the LEDs on the magnet monitor panel that correspond to Helium level, Nitrogen level and Helium Vessel Pressure start lighting up in cycles, with corresponding values for them being displayed on the display panel.
5	Disconnect the Laptop and cable from the Magnet Monitor	

IMPORTANT

If you need to reset, or power down the Magnet Monitor, do the following steps to properly shut down the software.

- Hold the *Sample* button in for about four seconds. **halt** should appear on the display.
- Release the *Sample* button for less than 1 second, and then press the *Sample* button again. **oooo** (four lower case o's) should appear on the display.
- Wait for at least **90 seconds**, then you may reset or power down the Magnet Monitor.

APPENDIX B - NULL MODEM DIAL-UP SETUP FOR LAPTOP

In order to properly connect your laptop, via the Null Modem cable, using the Dial-up Networking feature, you will need to install a “Modem” for this purpose. The modem to which I refer is not a hardware modem, although Windows will treat it that way.

IMPORTANT

If you need to reset, or power down the Magnet Monitor, do the following steps to properly shut down the software.

- Hold the *Sample* button in for about four seconds. **halt** should appear on the display.
- Release the *Sample* button for less than 1 second, and then press the *Sample* button again. **oooo** (four lower case o’s) should appear on the display.
- Wait for at least **90 seconds**, then you may reset or power down the Magnet Monitor.

B-1 Adding a Null Modem Connection

1. Open Start/Settings/Control Panel.
2. Select [Add new hardware].
3. Press [Next], to begin install wizard.
4. Press [No], to prevent Windows from searching for your modem.
5. Press [Next], to continue.
6. Select [Modem], to add a new modem.
7. Press [Next], to continue.
8. Select [Don’t detect my modem, I will select from list].

Note

If you do not see a Null Modem Type selection in the “Manufacturers” column, place your Magnet Monitor Proprietary software install disk in your laptop, and select “Have Disk” when prompted. A window should appear saying “copy manufactures files from” A:\. Select OK.

9. Select [Generic Null Modem] (or equivalent). The A drive should now be accessed.
10. Press [Next].
11. Select the COM port that will be attached to your null modem cable. You might have to select country, area code and dial out prefix.
12. Select an available COM port.

13. Press [Finish].

This ends the Modem install process.

Note

If you later have problems with this connection you may need to inspect the "Properties" of this new modem by viewing it in Start/Settings/Control Panel, open [Modems], find your "Null Modem" and check settings. All settings should be at the default values.

Now you'll need to create a new "Dial-up" connection in order to use this Modem.

B-2 Configuring old style Laptop

1. Open Start/Programs/Accessories/Dial-up Networking.

Note

If the above selection does not exist, add it to the Windows Settings found in Start/Settings/Control Panel, "Add/Remove Programs", "Windows Setup", "Communications". Verify that the "Dial-up Networking" box is selected.

2. In the "Dial-up Networking" window, select [Make New Connection].

3. In the "Select a Modem" box, select your Null Modem.

4. Type a Name for this connection, example would be Magnet Monitor.

5. Press [Configure].

The settings should be as follows;

- General
 - Communications Port (COM X), whatever you previously selected.
 - Maximum Speed, 115200. Do NOT select "Connect only at this speed".
- Connection
 - Use all default settings(8 data bits, no parity, 1 stop bit). Select Advance, use flow control, click on software.
- Options
 - Select [Bring up terminal window after dialing].
- All other settings leave as default.

6. Press [Next].

A phone number is required here, although this number is not actually used. Type in "521-0000". An area code is not needed.

7. Press [Next].

8. Press [Finish].

Your Null Modem connection is now ready for use.

Note

Refer to Appendix C, "Browser Setup", to learn how to use this connection to view the Magnet Monitor Status, Remote and Configuration screens.

B-2 Step 2 Configuring a DELL Laptop

1. Open Start/Programs/Accessories/Communications/Dial-up Networking.
2. In the “Dial-up Networking” window, select [Make New Connection].
3. In the “Select a Modem” box, select your Null Modem. Also type in a name for this connection such as “Magnet Monitor”. Then hit Next.
4. At the phone number prompt you must enter any seven digit number to continue. Then hit Next. Then hit **Finish** to complete the new connection.
7. To continue editing the connection, highlight the new connection by single clicking on it. Then select **File** which is located in the upper left hand section of the window, and select **Properties**.

Press [Configure].

The settings should be as follows;

- General
 - Communications Port (COM X), whatever you previously selected.
 - Maximum Speed, 115200. Do NOT select “Connect only at this speed”.
- Connection
 - Use all default settings(8 data bits, no parity, 1 stop bit). Select Advance, use flow control, click on hardware.
- Options
 - Select [Bring up terminal window after dialing].
- All other settings leave as default.

Press [Server Types].

- Have only the following 'checked'
- Enable software compression
- TCP/IP

All other boxes should be unchecked

5. Press **OK**.

Your Null Modem connection is now ready for use.

Note

Refer to Appendix C, “Browser Setup”, to learn how to use this connection to view the Magnet Monitor Status, Remote and Configuration screens.

APPENDIX C - BROWSER SETUP

Either Netscape Navigator or Microsoft Internet Explorer can be used as a browser. The setup is a little different for each.

C-1 Microsoft Internet Explorer

Note

It is recommended that you complete the Y2K upgrade on your PC prior to using Internet Explorer as the browser interface.

1. Open Start/Settings/Control Panel/Internet.
2. Open the tab "Connection".
3. Select [Connect to the Internet using a Modem].
4. Press [Settings].
5. Select the Dial-up connection you made for the Null Modem in appendix B.
6. Select OK to get out of the Dial-up Settings window, then select OK to get out of the Internet Properties window.
7. Open Start/programs/Accessories/Dial-up Networking. Then select the connection you made in Appendix B of this procedure.
8. Using the backspace key, remove any characters in the *Login* and *Password* fields.
9. Select the connect option.
10. You must get only the *login* prompt on the window that opened up automatically after selecting the connect option. If the login prompt is not present, try pressing the enter key twice. If the "status dialing" box remains on the screen until it times out, there is something wrong with the modem set-up. See table C-1 for possible causes
11. You may need to press the "enter" key to get the login prompt. For Login, type in **pppuser**.
12. For Password, type in **insite**.
13. You should then see characters scroll down the window. Select F7 to exit.
14. Open Internet Explorer. Shortly after Internet Explorer is opened select the **stop** button to halt any communication attempt. Then enter the internet address for the magnet monitor which is **10.0.0.5**

Note

If you do not get a response at this point you might need to reload Internet Explorer. Refer to table C-1 for instructions.

C-2 Netscape Navigator 2.02

1. Open Netscape Navigator.
2. In menu, open Options/Network Preferences.
3. Select [No Proxies].
4. Select [OK].

C-3 Netscape Navigator 4.01

1. Open Netscape Navigator.
2. In menu, open Edit/Preferences.
3. Expand the “Advanced” selection.
4. Select [Proxies].
5. Select [Direct connection to the Internet].
6. Select [OK].

C-4 Connecting your Browser to the Magnet Monitor

1. Using a Null Modem cable, connect your laptop to the Magnet Monitor. Use the “Laptop” connector on the Magnet Monitor.
2. Make sure the Magnet Monitor is powered up. Hit the reset switch on the magnet monitor. It will take 2 minutes for the magnet monitor to boot up.
3. Find your Dial-up for the Null Modem connection, activate it.
When the connection is made, you will get a pop up window.
4. If the terminal window is blank, press [Enter] twice. A prompt should appear `Login`.

Note

If this attempt to login fails, re-boot the MM and repeat from C-4, step 1.

5. Type in **pppuser**, press [Enter].
6. Type in **insite** at the password prompt.
7. A string of characters will show up. Press F7 to continue.
8. Open your browser.

9. Whichever browser you've selected, type in the URL IP address **10.0.0.5** and activate the search.
10. Momentarily you should see the Magnet Monitor screen come up. If you do not see this, check all your settings and connections.

TABLE C-1
Troubleshooting tips

Problem	Possible cause
You cannot connect to the magnet monitor, after selecting the icon in Dial-up Networking. The laptop and magnet monitor will not establish communication.	In the dial-up networking box, highlight the connection icon by single clicking on it. Then hit the right mouse button and select properties. Select <i>configure</i> , then <i>options</i> . Make sure "bring up a terminal window after dialing" is selected.
The option "connect to the internet using a modem" cannot be selected and you have not done the Y2K upgrade. To verify go into Internet Explorer, select the <i>view</i> pull-down menu, Select <i>Internet options</i> , then <i>connections</i> . You must be able to select the option "Connect to the internet using a modem".	Internet Explorer software will need to be reloaded using the following procedure. You must have the Blue Disaster Recovery disk dated 1999 to do this procedure. 1. Go to My Computer/Control Panel/Add Remove Programs. 2. Highlight Microsoft Internet Explorer 4.0. 3. Select Add/Remove, highlight un-install Internet Explorer 4.0 and all its components, and select OK. If asked, DO NOT remove any personal files or links You may have to select OK two more times before the program files are removed. You must close all programs if prompted. It takes at about 7 minutes for all files to be removed. 4. Reboot your laptop. 5. Go to the D drive or CD-ROM, select Add-ons, ie4,lesetup.exe. 6. Go to Explorer/Support/Internet Explorer 4.01/lesetup.exe 7. Run setup. 8. Re-install ie4. It takes about 17 minutes for Internet Explorer to install. 9. Follow the procedure to connect to magnet monitor.

IMPORTANT

If you need to reset, or power down the Magnet Monitor, do the following steps to properly shut down the software.

- Hold the *Sample* button in for about four seconds. **halt** should appear on the display.
- Release the *Sample* button for less than 1 second, and then press the *Sample* button again. **oooo** (four lower case o's) should appear on the display.
- Wait for at least **90 seconds**, then you may reset or power down the Magnet Monitor.

APPENDIX D - COEF.CFG FILE

The following listing is a copy of the coef.cfg files used in the Magnet Monitor for the purpose of converting measured voltages into the displayed measured values with appropriate units. Refer to Appendix E for information on how to use these coefficients.

```
# COEFDATA.CFG -- Coefficient Data File Rev. 1.02 10 Sept, 1998..Modified He Level
coefficients
# Coefficient Data File Rev. 1.01 9 Sept, 1998...modified RuO data for
gain 65*2 and 1.5K to 24K range, see "New_limited_25.mcd"
# and
"Vendor_Data_RuO.xls". See "coef_9_10_98.cfg"
# Coefficient Data File Rev. 1.00 22 July, 1998..see "Coef_9_98.cfg for
this version
# Based on the EDM file coefdata.cfg Rev. 3.0 May 15, 1995
# Added tables for SI-410 Diode and Ruthinium Oxide Diode.
#
# Curve #0 A to D Tics
# Used for debugging
0 0 0 4095 0.000 1.000E+3 0.000 0.000 0.000 0.000
#
# English Units
#
# Curve #1 0-5 scale
# Used for He Flow and Water Flow (and Voltage for debugging)
# Units: Use "SCFH" for He Flow, and "GPM" for water flow.
1 0 0 4095 0.000 1.221001 0.000 0.000 0.000 0.000
#
# Curve #2 0-100 scale
# Used for Water Temp
# Units: Use "DEG F"
2 0 0 4095 0.000 2.442002E+1 0.000 0.000 0.000 0.000
#
# Curve #3 0-100 scale
# Used for He Level
# Units: Use "%"
3 0 0 4095 1.000E+2 -2.442002E+1 0.000 0.000 0.000 0.000
#
# Curve #4 0-15 scale
# Used for He Pressure
# Units: Use "PSIG"
4 0 0 819 0.000 0.000 0.000 0.000 0.000 0.000
4 1 820 4095 -3.755725 4.580153 0.000 0.000 0.000 0.000
#
# Curve #5
# Used for DT500D Diode Temp
# Units: Use "K"
5 0 0 1406 +4.457076E+2 -2.277978E+2 0.000 0.000 0.000 0.000
5 1 1407 1685 +5.929520E+4 -1.917673E+5 +2.492882E+5 -1.621930E+5 +5.276006E+4 -6.865506E+3
5 2 1686 1920 -1.775261E+5 +3.343076E+5 -1.927495E+5 +7.476784E+3 +2.547142E+4 -5.857463E+3
5 3 1921 3467 +6.370002E+2 -1.073492E+3 +7.577936E+2 -2.691138E+2 +4.774339E+1 -3.389978
5 4 3468 4095 +4.917566E+1 -1.158421E+1 0.000 0.000 0.000 0.000
#
# Metric Units
#
# Curve #6
# Used for Water Flow
# Units: Use "LPM"
6 0 0 4095 0.000 4.621978 0.000 0.000 0.000 0.000
#
# Curve #7
# Used for Water Temp
# Units: Use "DEG C"
7 0 0 4095 -1.777778E+1 1.356668E+1 0.000 0.000 0.000 0.000
#
# Curve #8
# Used for He Pressure
# Units: Use "KPA"
```

```

8 0 0 819 0.000 0.000 0.000 0.000 0.000 0.000
8 1 820 4095 -2.587269E+1 3.155206E+1 0.000 0.000 0.000 0.000
#
# Curve #9
# Used for He Flow
# Units: Use "SLPM"
9 0 4095 0.000 5.762320E-1 0.000 0.000 0.000 0.000
#
# Curve #10
# Spare
10 0 0 4095 0.000 1.000E+3 0.000 0.000 0.000 0.000
#
# Curve #11
# Used for SI410 Diode Temp
# Units: Use "K"
11 0 0000 1450 +5.919119E+2 -3.044092E+2 0.000 0.000 0.000 0.000
11 1 1451 1778 +2.843001E+4 -8.599055E+4 +1.051892E+5 -6.441759E+4 +1.969781E+4 -2.409274E+3
11 2 1779 1926 -5.637970E+5 +9.241173E+5 -3.167606E+5 -2.145678E+5 +1.664921E+5 -3.014900E+4
11 3 1927 2816 +1.938138E+2 -6.084114E+2 +8.611591E+2 -5.592035E+2 +1.665666E+2 -1.860774E+1
11 4 2817 4096 0.000 0.000 0.000 0.000 0.000 0.000
#

# Curve #12
# Used for Ruthinium Oxide Diode Temp
# Units: Use "K"
12 0 0 1163 +24.00 0.000 0.000 0.000 0.000 0.000
12 1 1164 1194 +9.982059776E+5 -3.3539329599E+6 +4.2284751009E+6 -2.3706324942E+6 +4.9863932411E+5
0.000
12 2 1195 1216 -9.1044797833E+5 +3.0422111121E+6 -3.8102628057E+6 +2.1201554523E+6 -4.4224511757E+5
0.000
12 3 1217 1250 +4.6729189182E+6 -1.5156947805E+7 +1.8436078306E+7 -9.9664653362E+6 +2.0204163527E+6
0.000
12 4 1251 1308 -1.3433797285E+4 +4.5357130871E+4 -5.6691383489E+4 +3.1200157128E+4 -6.3934947192E+3
0.000
12 5 1309 1933 +9.0258123802E+2 -2.054783153E+3 +1.7657833504E+3 -6.7514891513E+2 +9.6648164149E+1
0.000
12 6 1934 4095 +1.500 0.000 0.000 0.000 0.000 0.000

```

APPENDIX E - CALCULATING WITH THE COEF.CFG FILE DATA

The MM utilizes multiple polynomials to facilitate curve fitting of the measured voltage to the displayed value. Some sensors have simple straight line transfer functions, others are non-linear curves. Each measured parameter can have up to 5 separate fifth order polynomials to characterize different ranges of the value. In the cases where a straight line is all that is required, a single polynomial is used, with only the first 2 coefficients non-zero. In more complex functions, all 5 polynomials may be used, each having up to 5 separate coefficients. Refer to Appendix D Coef.cfg file, to see a copy of the coefficient table used in the MM. The table below is a copy of one of these coefficient sections. The one shown is for the RuO sensors, Coefficient table 12.

The subsequent equations demonstrate how to use the coefficient tables to calculate what a displayed value would be, given a measured voltage from an RuO sensor.

The actual RuO sensors produce an output voltage of approximately 10 to 30 millivolts. Before making it to the MM, this signal is amplified by a gain of 65 in the RuO Buffer amplifier (Located at the magnet). On entry into the MM, the signal is further multiplied by 2.

The MM routes this signal to an Analog-to-Digital converter (ADC). This converter is a 12 bit conversion process. The ADC is scaled to read uni-polar 0 to 5 Volts. The following equations use ADC counts (0 to 4095), rather than actual voltage. Another important fact to keep in mind, the MM software also divides this count by 1000 prior to applying a value to the polynomials. Therefore, as an example, if you read 2.5 volts with an external meter, the number you would plug into the equations (all coefficient tables are handled the same way) would be as follows:

$$V_{\text{measured}} := 2.5$$

$$AD_counts := 2.5 \cdot \frac{2^{12}}{5} \quad \text{therefore...} \quad AD_counts = 2048$$

$$\text{Equation_input} := \frac{AD_counts}{1000} \quad \text{therefore...} \quad \text{Equation_input} = 2.048$$

Keep this in mind when experimenting with the equations below...

Curve #	Coef Set	A/D Count reading		Coefficients					
		Lower	Upper	Coef ₀	Coef ₁	Coef ₂	Coef ₃	Coef ₄	Coef ₅
12	0	0	1163	24	0	0	0	0	0
12	1	1163	1194	9.98206E+5	-3.35393E+6	+4.22848E+6	-2.37063E+6	+4.98639E+5	0.000
12	2	1194	1216	-9.10448E+5	+3.04221E+6	-3.81026E+6	+2.12016E+6	-4.42245E+5	0.000
12	3	1216	1250	+4.67292E+6	-1.51569E+7	+1.84361E+7	-9.96647E+6	+2.02042E+6	0.000
12	4	1250	1308	-1.34338E+4	+4.53571E+4	-5.66914E+4	+3.12002E+4	-6.39349E+3	0.000
12	5	1308	1933	+9.02581E+2	-2.05478E+3	+1.76578E+3	-6.75149E+2	+9.66482E+1	0.000
12	6	1933	4095	+1.500	0.000	0.000	0.000	0.000	0.000

This demonstration will be done using only Coef set #5, from the above table. The equation using this set of coefficients can only be used for voltage readings that result in ADC counts in the range of 1308 to 1933. This correlates to a voltage range as shown below, as would be measured at the RuO sensor.

$$V_{lower} := 1308 \cdot \frac{5}{2^{12}} \cdot \frac{1}{65.2} \quad \text{therefore...} \quad V_{lower} = 0.012$$

$$V_{upper} := 1933 \cdot \frac{5}{2^{12}} \cdot \frac{1}{65.2} \quad \text{therefore...} \quad V_{upper} = 0.018$$

For the demonstration, we first need to create our "Input Voltage", for simplicity I'll assume you can do that calculation and I'll just create a number that ranges from 1308 to 1933, that variable is AD.

$$x := 0, 1.. (1933 - 1308)$$

$$AD_x := x + 1308$$

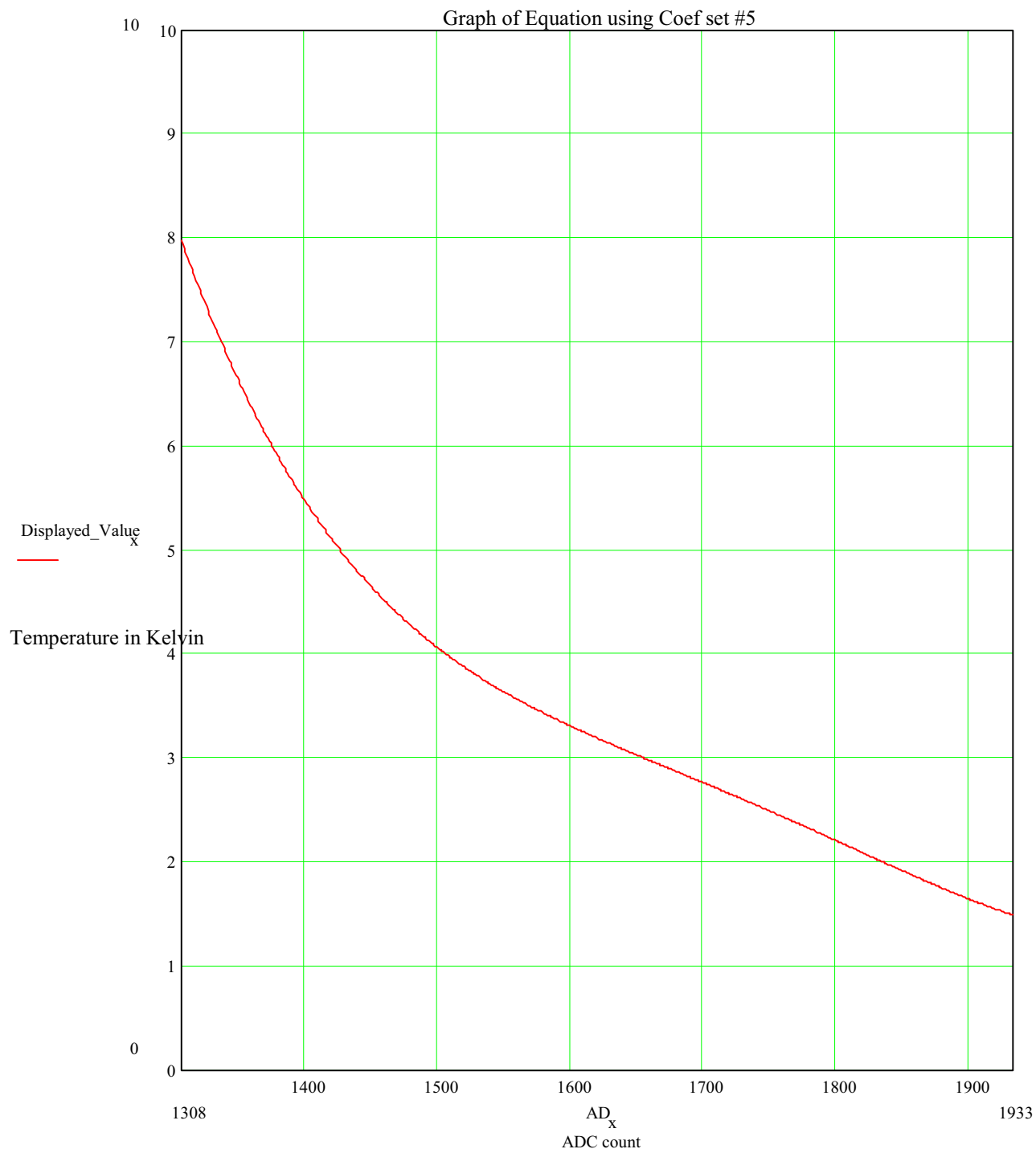
Next define the coefficients (taken from the table above)...

...and now perform the calculation for all values of "x" (i.e. 1308 to 1933)...

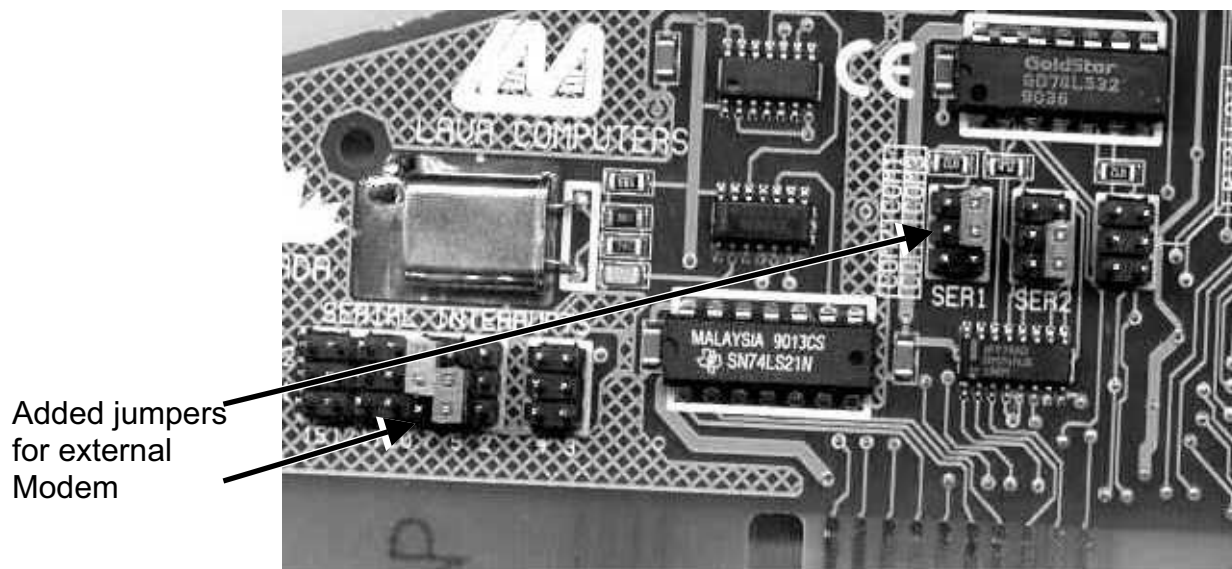
$$Coef_0 := 9.02581 \cdot 10^2 \quad Coef_1 := -2.05478 \cdot 10^3 \quad Coef_2 := 1.76578 \cdot 10^3 \quad Coef_3 := -6.75149 \cdot 10^2 \quad Coef_4 := 9.66482 \cdot 10^1 \quad Coef_5 := 0$$

$$Displayed_Value_x := Coef_0 + Coef_1 \cdot \frac{AD_x}{1000} + Coef_2 \cdot \left(\frac{AD_x}{1000} \right)^2 + Coef_3 \cdot \left(\frac{AD_x}{1000} \right)^3 + Coef_4 \cdot \left(\frac{AD_x}{1000} \right)^4 + Coef_5 \cdot \left(\frac{AD_x}{1000} \right)^5$$

The following graph demonstrates the computed value "Displayed Value", given the various values of AD...

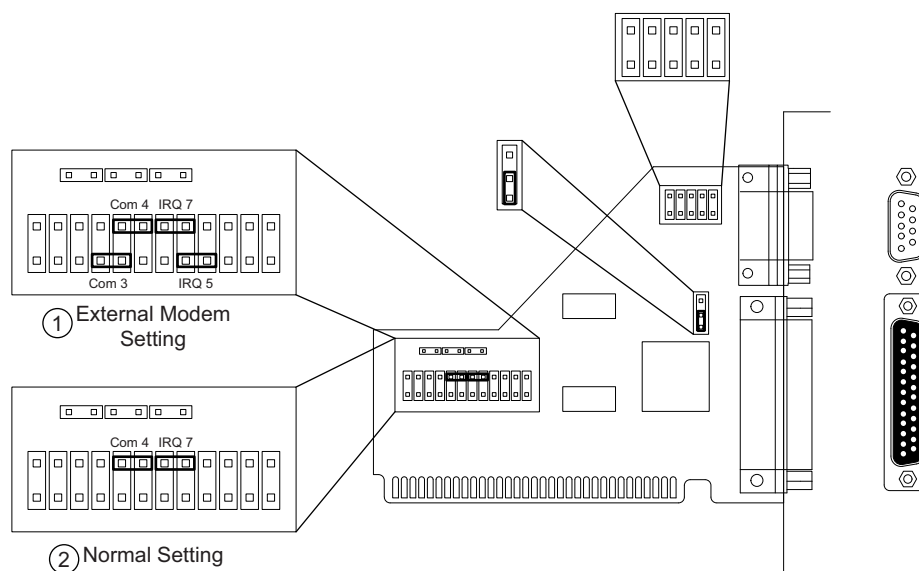


APPENDIX F - GENERAL I/O BOARD JUMPER SETTINGS FOR LATER GRANITE (POST JANUARY 2001) UNITS



GENERAL I/O BOARD
ILLUSTRATION F-1

GENERAL I/O BOARD JUMPER SETTINGS FOR EARLY GRANITE (PRE JANUARY 2001) UNITS



GENERAL I/O BOARD
ILLUSTRATION F-2

APPENDIX G - PPADSCMD EMAIL

G-1 Overview

The following sections provide the procedures necessary to receive Magnet Monitoring information by sending email to PPADSCMD.

- SECTION G-2, RECEIVING MAGNET MONITORING DATA THROUGH EMAIL - This section provides the following information.
 - How to address email to PPADSCMD from the various types of GE email accounts.
 - What the general guidelines are for sending email to PPADSCMD.
 - How to use the online help files to find the correct command syntax.
 - When to use the BEGIN and END commands to prevent PPADSCMD from incorrectly processing information embedded in the email message.
- SECTION G-3, FINDING SITE SPECIFIC INFORMATION - This section describes some common combinations of commands for keeping track of what is happening (alarms and fills) at specific sites and how to begin troubleshooting problems at specific sites with Magnet Monitoring.
- SECTION G-4, GETTING INFORMATION ABOUT ALL MAGNET MONITORING SITES - This section describes some common combinations of commands for checking the revisions, connectivity, and settings at all of the sites with Magnet Monitoring.
- SECTION G-5, RECEIVING MAGNET MONITORING PROGRAM INFORMATION - This section provides procedures for receiving current Magnet Monitoring information, including Magnet Monitoring part numbers, toll free phone numbers, and computer usage. You can also address specific questions to the PPADS administrator.
- SECTION G-6, ENTERING AND RETRIEVING TEAM INFORMATION - This section gives the list of commands to use to enter your contact information, join the email distribution list for a team, and find the information to contact another member of a team.
- SECTION G-7, PPADSCMD COMMAND SUMMARY - This section provides an alphabetical list of the available types of PPADS commands.

G-2 Receiving Magnet Monitoring Data Through Email

To access Magnet Monitoring information, you need the following.

- To have a GE email account: CC:Mail, Macintosh/Microsoft Mail, Mainframe (Wiz Mail), Microsoft Exchange Mail, or UNIX.
- To meet the PPADSCMD guidelines.
- To know the correct command syntax. For example, you must provide the information inside the angle brackets, <>, (such as, LIST DISPATCHES <System ID>); you may optionally provide the information in the square brackets, [], (such as, REPORT [SHORT or LONG]). HELP for the specific commands (for example, HELP SHOW) provides the most current information on command syntax.
- To use the BEGIN and END commands, when it is necessary to stop the PPADS System from trying to process any header or footer information embedded in your email messages.

G-2-1 Address Email

To address email to PPADSCMD, use one of the addresses in Table G-1. Note that only Mainframe/Wizz mail requires the Subject field be completed. PPADS does not process the Subject field; however, if the subject field is completed (in any of the GE email systems), PPADS' reply identifies the entry in the Subject field.

TABLE G-1
ADDRESSES FOR EMAIL TO PPADSCMD

EMAIL TYPE	EMAIL ADDRESS
CC:Mail/Laptop	To: PPADSCMD
Macintosh/ Microsoft mail	To: ppadscmd@iscmed.med.ge.com
Mainframe/Wiz mail	To: PPADSCMD Subject: XXX (NOTE: This field must be filled.)
Microsoft exchange mail ¹	To: ppadscmd@iscmed.med.ge.com
PROFS mail ¹	To: ppadscmd (NOTE: Use the GEMS-Am email directory, User ID: I000216)
UNIX/Internet	To: ppadscmd@iscmed.med.ge.com
YMSNET	To: ppadscmd
¹ To send commands for Magnet Monitoring information from Microsoft Exchange Mail and PROFS, start the list of PPADS commands with the BEGIN command and stop the list with the END command. Microsoft Exchange Mail and PROFs place information in the body of the message that can cause an error unless the BEGIN and END commands are used.	

G-2-2 PPADSCMD Guidelines

Use the following guidelines to access Magnet Monitoring data.

- Place all commands to the database in the body of the email message.
- Place only commands in the body of a PPADSCMD email message. The PPADS system expects only commands in the body of a PPADSCMD Email message, so any other information causes an error. If the PPADS list server does not recognize a command, all following commands are ignored.
- Keep each command on its own line.
- Send email to PPADSCMD with the NEWS command every couple of months. Additional commands are available occasionally. Also, use the HELP command periodically to see an overview of all commands.

G-2-3 Use the HELP Command to See Correct Command Syntax

The HELP command without an argument sends the most current version of the general guidelines for sending email to PPADSCMD. To get detailed “help” on a specific command, use the command name as the argument.

1. Address email to PPADSCMD (Refer to Section G-2-1, Address Email.)
2. Place the HELP [ADMIN, BEGIN, END, LIST, NEWS, REPORT, SET, SHOW, SUBSCRIBE, or UNSUBSCRIBE] command on a new line in the body of the message.
3. Send the message. See Illustration G-1.

Command: HELP ADMIN

Reply: -----PPADS EMAIL COMMAND PROCESSOR RESULTS

Command: help admin

Help for the PPADS ADMIN Command
PPADS Information Server
Magnet Monitoring
MR Global Readiness
Advanced Service Methods

WARNING

Permission for the use of this software by persons other than GE Medical Systems employees is NOT GRANTED. A license to use Advanced Service Material does NOT extend to or cover this material. Also, a license to use Advanced Service Material does NOT extend to or cover this material.

If you are not a GE Medical Systems employee, delete this document immediately.

ADMIN COMMAND CAPABILITIES

The ADMIN command sends a copy of your PPADSCMD Email to the PPADS administrator. When the PPADS system recognizes the ADMIN command, the system ignores all the text after the command. Use ADMIN to send messages to the administrator.

USING THE PPADS ADMIN COMMAND

Use the following procedure to send messages to the PPADS administrator.

- 1) Address e mail to PPADSCMD. (Refer to the main PPADSCMD Help for information on addressing e mail messages.
- 2) Place the ADMIN command on a new line in the body of a new e mail message; then enter the message for the PPADS administrator. (Place no other commands in a message with the ADMIN command.)
- 3) Send the message.

Example: admin

For the last week, I haven't been getting e mail alarm notices for the sites that I'm subscribed to. Could you check on it? Thanks, Kate

Example Reply: Sent to the list server administrator!

EXAMPLE HELP COMMAND ILLUSTRATION G-1

G-2-4 Use the BEGIN and END Commands to Prevent Incorrect Processing

The BEGIN command marks the beginning of a list of PPADS commands. Use this command to prevent PPADS from trying to process header information that is embedded in the message. Use this command if you are entering commands (email) from Microsoft Exchange Mail or from PROFS.

The END command marks the end of your PPADS commands. Use this command to prevent PPADS from trying to process footer information that is embedded in the message. Use this command if you are entering commands (mail) from Microsoft Exchange Mail or PROFS or if you use a .signature.

1. Address email to PPADSCMD (Refer to Section G-2-1, Address Email.)
2. Place the BEGIN command on a new line in the body of the message before the other command(s).
3. Place the END command on a new line after the other PPADS command(s).
4. Send the message. See Illustration G-2.

```

Command:  begin
          show db 219879 mrc
          end

Reply:    ----- PPADS EMAIL COMMAND PROCESSOR RESULTS -----

          Command:  show db 219879mrc

          System ID:  219879MRC
          ST ANTHONY HOSPITAL/CRS
          Customer:   390820446
          MTL RUBEN GAITAN, XXXX, BALZER, 5x, MUX=4
          Class:     nam
          Country:   usa
          Room ID:   CRYOGENS
          System Type:  SIGNA34
          EDM REV:   4.50
          EDM Phone:  2198771522w44
          Customer Phone: (219)874-0520
          Primary FE:  LEE, JAMES, E
          Magnet Type:  SX
          Magnet S/N:  XXXX

          Individuals subscribed to site:

          Last Contact:  Tue Nov 19 05:20:24 CST 1996
          Last Alarm:   Thu Oct 24 13:09:09 CST 1996
          Last Fill:    Mon Aug 12 20:59:26 CST 1996
          Last Checkin: Tue Nov 19 05:20:23 CST 1996
          Last Fetch:   Mon Jun 17 20:56:22 CDT 1996
          Last Test:    Wed Oct 11 17:31:23 CDT 1995
  
```

EXAMPLE BEGIN AND END COMMANDS
 ILLUSTRATION G-2

G-3 Finding Site Specific Information

PPADS offers a great deal of site specific information. For a complete list of applicable commands, send email to PPADSCMD with the HELP SHOW and HELP SUBSCRIBE commands.

If you are responsible for a specific site or a group of sites, you may want to do the following tasks:

- To receive email when a specific site has an alarm or fill , send email to PPADSCMD with the following command. (See Illustration G-3.)
 SUBSCRIBE EMAIL <system ID>

Command: subscribe 215431CRY

Reply: ----- PPADS EMAIL COMMAND PROCESSOR RESULTS -----

Command: subscribe email 215431cry

You have been added to the Email list for 215431CRY
 You will be notified by email in the event of an alarm or fill at
 this site.

In the event that you wish to be removed from this site's list,
 send an unsubscribe command to this list server.

EXAMPLE SUBSCRIBE EMAIL COMMAND ILLUSTRATION G-3

- To check that a specific site is able to connect to PPADS, send email to PPADSCMD with the following command:
 SHOW CALLS <system ID> 15

Note that the "15" in the previous command refers to the number of lines to display from the call.log.

- To view the most current parameter values at the site, send email to PPADSCMD with the following command. (See Illustration G-4.)
 SHOW CURRENT <system ID>

G-3 Finding Site Specific Information (continued)

Command: show current 209435cryo2

Reply: ----- PPADS EMAIL COMMAND PROCESSOR RESULTS -----

Command: show current 209435cryo2

The last data to be reported to the PPADS sytem was on_97/04/09!10:22

The He level was 88.8

The 1st stage temp was 47.4

The 2nd stage temp was 11.4

The He pressure was 0.93

The water flow rate was 3.04

The inlet water temp was 57.09

**EXAMPLE SHOW CURRENT COMMAND
ILLUSTRATION G-4**

- To see the site's name, location, telephone numbers, primary field engineer, time of last contact with PPADS, and who is subscribed to receive email when the site has an alarm or fill, send email to PPADSCMD, with the following command:
SHOW DB <system ID>
- To troubleshoot a problem at a site, send email to PPADSCMD with one or more of the following commands. Note that you may need to see the results of a message before sending the other commands. For an example of the LIST DISPATCHES command, see Illustration G-5.
LIST DISPATCHES <system ID>
SHOW DISPATCH <system ID> <dispatch #>
SHOW CURRENT <system ID>
SHOW MAGALARM <system ID> 25
SHOW EDM <system ID> 25
SHOW MAGDATA <system ID>

Note that the you can import the data from the SHOW MAGDATA into spreadsheet software.

Note that the "25" in the previous commands refers to the number of lines to display from either the magalarm.log or the edm.log.

G-3 Finding Site Specific Information (continued)

Command: list dispatches 214247rmrc

Reply: ----- PPADS EMAIL COMMAND PROCESSOR RESULTS -----

Command: list dispatches 214247rmrc

Mar	24	1995	DISP0270175003
Mar	24	1995	DISP0270178335
Mar	24	1995	DISP0270178755
Mar	24	1995	DISP0270180556
Mar	24	1995	DISP0270183897
Mar	24	1995	DISP0270174710
Mar	24	1995	DISP0270174965
Mar	24	1995	DISP0270183627
Mar	24	1995	DISP0270183705
Sep	18	07:44	DISP0270206439

EXAMPLE LIST COMMAND ILLUSTRATION G-5

G-4 Receiving Information About All Magnet Monitoring Sites

PPADS offers information about the sites with Magnet Monitoring. For a complete list of applicable commands, send email to PPADSCMD with the HELP COUNT, HELP LIST, and HELP REPORT commands.

If you are responsible for a group of sites, you may want to do the following tasks:

- To see the total number of sites with Magnet Monitoring divided into geographic location, send email to PPADSCMD with the following command. (See Illustration G-6.)

REPORT SHORT

Command: report short

Reply: ----- PPADS EMAIL COMMAND PROCESSOR RESULTS -----

Command: report short

*** PPADS Usage ***

Total number of registered sites: 868

Total number of dispatches: 23115

Total number of calls: 97343

Registered American Sites: 797

Registered European Sites: 48

Registered Asian Sites: 5

Registered Australian Sites: 2

***PPADSCMD EMAIL SERVER USAGE ***

Number of Users: 404

Number of Command Requests: 18392

EXAMPLE REPORT SHORT COMMAND
 ILLUSTRATION G-6

G-4 Receiving Information About All Magnet Monitoring Sites (continued)

- To check how many sites have been able to connect to PPADS recently (in this case, within 15 days), send email to PPADSCMD with the following command (see Illustration G-7): **COUNT FRESH 15 CONTACT**

Command: count fresh 99 check
count fresh 99 test

Reply: ----- PPADS EMAIL COMMAND PROCESSOR RESULTS -----

Command: count fresh 99 check

314 sites have had a check in last 99 days

Command: count fresh 99 test
201 sites have had a test in last 99 days

EXAMPLE COUNT FRESH COMMAND ILLUSTRATION G-7

G-4 Receiving Information About All Magnet Monitoring Sites (continued)

- To check how many sites have been unable to connect to PPADS recently (in this case, 45 days), send email to PPADSCMD with the following command:
COUNT STALE 45 CONTACT

If you want to troubleshoot why these sites are unable to connect, first list the system IDs of these sites. Send email to PPADSCMD with the following command:

LIST STALE 45 CONTACT

Alternatively, you may want to print a list of each site with Magnet Monitoring in order according to the last date and time that the sites had contact with PPADS. To see a list of the sites according to their last contact with PPADS, send email to PPADSCMD with the following command:

LIST LAST CONTACT DATE

Once you have the list of system IDs, send email to PPADSCMD with the following command for each of the system IDs. (Note that as long as you put each of these commands on a new line, they do not have to be in different messages.)
SHOW DB <system ID>

For example,

SHOW DB 1073CRYO

SHOW DB 1100CRYO

SHOW DB 1196CRYO

SHOW DB 1325CRYO

etc.

Now that you have the modem telephone number for each of these sites, you can try to connect directly with the site and check its .log files, messages from field engineers (these would be left in the Note fields of the Error Reporting Setup screen), and the site's modem configuration file.

- To monitor sites that may have lost connectivity recently (in this case in the last 10 days), send email to PPADSCMD with the following command:
LIST DIFF FRESH 10 CONTACT FRESH 20 CONTACT
- To list the sites according to MAC (Magnets And Cryogens) Team Leader, if the sites are in North America, send email to PPADSCMD with the following command:
LIST MTL

Note that the COUNT MTL command just gives the number of sites assigned to each MAC Team Leader, if applicable.

- To generate a list of every site with Magnet Monitoring, send email to PPADSCMD with the following command:
LIST SITES

G-5 Receiving Magnet Monitoring Program Information

PPADS offers information about the Magnet Monitoring program. For a complete list of applicable commands, send email to PPADSCMD with the HELP ADMIN, HELP LIST, HELP SUBSCRIBE, HELP NEWS, and HELP REPORT commands.

If you are installing Magnet Monitoring equipment at new sites or are working with existing sites, you may want to do the following tasks:

- To ask questions of the PPADS administrator or notify him of possible problems, send email to PPADSCMD with the following command (see Illustration J-8):
ADMIN <your message>

Command: admin

Could you please send me a copy of the current MRT pager list? Chris

Reply: Sent to the list server administrator!

EXAMPLE ADMIN COMMAND ILLUSTRATION G-8

- To see a list of all the email distributions (and their purposes), send email to PPADSCMD with the following command:
LIST DISTRIBUTIONS
- To be notified by email when information becomes available that pertains to a specific distributions, send email to PPADSCMD with the following command:
SUBSCRIBE DISTRIBUTION <distribution name>
- To list all of the available Magnet Monitoring parts and tools, send email to PPADSCMD with the following command:
NEWS PARTS
- To get the current list of international toll free numbers (for EDMs to call PPADS), send email to PPADSCMD with the following command:
NEWS PHONES
- To receive information on registering unusual sites (such as, MRT sites), send email to PPADSCMD with the following command:
NEWS REGISTRATION
- To see site statistics, email server usage, the time of the last calls to the PPADS machines, the time and date that the PPADS machines were last started, the status of the PPADS machines, and statistics about processes and data storage on the PPADS machines, send email to PPADSCMD with the following command:
REPORT LONG
- To see the number of alarms waiting to have dispatches cut, send email to PPADSCMD with the following command:
COUNT ALARMQUEUE

G-6 Entering And Retrieving Team Contact Information

PPADS offers information about entering and retrieving team information. For a complete list of applicable commands, send email to PPADSCMD with the HELP LIST, HELP SHOW, HELP SET, HELP SUBSCRIBE, and HELP UNSUBSCRIBE commands.

These tasks let you either enter information into a Magnet Monitoring “address book” or check existing information in that address book. If you want to join or quit teams or view the contact information for a team member, you may want to do the following tasks:

- To see the exact names of the teams, send email to PPADSCMD with the following command: **LIST TEAMS**
- To put your contact information into PPADS and join one or more teams, send email to PPADSCMD with the following commands. (Note that you can set additional information; refer to the HELP SET command.)
SET FIRSTNAME <your first name>
SET LASTNAME <your last name>
SET ADDRESS <your address>
SET ASPEN <your 5-digit Aspen mailbox number>
SET CPHONE <your cellular phone number>
SET OPHONE <your office phone number>
SUBSCRIBE TEAM <team name>
- To view the contact information of all the persons on a team, send email to PPADSCMD with the following command: **SHOW TEAM <team name>**
- To remove your contact information from a team list, send email to PPADSCMD with the following command: **UNSUBSCRIBE TEAM <team name>**

G-7 PPADSCMD Command Summary

Use PPADS commands to receive information about Magnet Monitoring. You must provide the information inside the angle brackets, <>, (for example, <Dispatch#>); you may optionally provide the information in the square brackets, [], (for example, HELP [ADMIN, BEGIN, END, LIST, NEWS, REPORT, SET, SHOW, SUBSCRIBE, or UNSUBSCRIBE]).

The following commands are used most frequently.

- ADMIN - Use to notify of problems or send requests to the PPADS administrator.
- BEGIN and END - Use to prevent PPADSCMD from trying to process embedded header or footer information. Always use these commands when sending mail from Microsoft Exchange Mail or from PROFS.
- COUNT - Use to see the number of number of dispatches at a site; the number of Magnet Monitoring sites per MAC Team Leader; or the number of alarms, checks, contacts, fills, or tests within or before a specified number of days.
- HELP - Use to see the most recent commands and general guidelines for using those commands. To see specific help on an individual command, send the HELP <command type> message to PPADSCMD. For example, the HELP SHOW command shows you how to use each of the SHOW commands and provides example responses for each command (such as, SHOW CONFIG, SHOW DB, SHOW EDM, SHOW MAGALARM, and SHOW MAGDATA).
- LIST - Use to list all the Magnet Monitoring sites; all the dispatches for a specific site; all the sites according to MAC Team Leader; the sites that have had alarms, checks, contacts, fills, or tests within or before a specified number of days; or the sites' last alarms, checks, contacts, fills, or tests either according to cryogen system ID or by date .
- NEWS - Use to see the most recent toll free numbers, the most recent part numbers, or the most recent software information.
- REPORT - Use to get statistics on both the Magnet Monitoring sites and the PPADS System.
- SET - Use to provide contact information prior to subscribing to a team.
- SHOW - Use to see a site's EDM and magnet configurations, contact information, status of a specific dispatch, EDM log, magnet alarm log, or magnet statistics.
- SUBSCRIBE and UNSUBSCRIBE - Use to place or remove your email address on a list that causes you to receive email when the site you "subscribed" to has an alarm or fill.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	Sept 4, 2000	P. Kargard	Initial Release
1	April 2, 2001	P. Kargard	Updates for release 2.0 software
2	October, 2001	P. Kargard	New software load procedure
3	Oct, 2003	P. Kargard	Upgrade for revision 2.3 software